



Department of Electrical and Computer Engineering

North South University

Directed Research

Heart Failure Prediction Using Machine Learning

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LETTER OF TRANSMITTAL

December, 2024

To

Dr. Mohammad Abdul Matin

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Project Report on “Heart Failure Prediction Using Machine Learning”

Dear Sir,

With due respect, we would like to submit our **CSE498R Directed Research** report on “**Heart Failure Prediction Using Machine Learning**” for the completion of our CSE498R as required by the Department of Electrical and Computer Engineering, North South University. The report outlines our study aimed at creating a robust machine learning system capable of predicting the likelihood of heart failure by leveraging a comprehensive dataset. This project provided us with invaluable experience in applying advanced data analysis and machine learning techniques to address a critical healthcare challenge. It also enhanced our research skills, offering insights into real-world applications of predictive analytics in healthcare. With sincerity and dedication, we have strived to meet all the requirements and maintain the quality expected for this report.

We will be highly obliged if you receive this report and provide your valuable judgment. It would be our immense pleasure if you found this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

.....

Shishir Oyshi 2014334642

ECE Department

North South University, Bangladesh

.....

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APPROVAL

Shishir Oyshi (ID 2014334642) and Tohura Ismail Peea (ID 2012340042) from the Electrical and Computer Engineering Department of North South University have worked on the Senior Design Project titled “**Heart Failure Prediction Using Machine Learning**” under the supervision of Dr. Md. Alamin partial fulfillment of the requirement for the degree of Bachelor of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

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Chairman’s Signature

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Dr. Mohammad Abdul Matin

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Department of Electrical and Computer Engineering

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere, partially or entirely, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

1. Shishir Oyshi

2. Tohura Ismail Peea

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Furthermore, we would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for facilitating the research. We would like to thank our loved ones for their countless sacrifices and continual support.

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ABSTRACT

Heart disease continues to be a principal cause of death globally, especially in underdeveloped areas, underscoring the essential requirement for early identification and preventive strategies. This study employs machine learning techniques to predict heart failure, leveraging a comprehensive dataset encompassing 23 clinical variables such as age, sex, family history, and hyperlipidemia. The methodology includes data preprocessing techniques aimed at addressing missing values and class imbalance, feature engineering to identify complex interactions, and the application of eight predictive models: Convolutional Neural Networks (CNNs), TabNet, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), Gaussian Naive Bayes, Bagging Decision Trees, and Voting Classifiers. CNN outperformed the other models, achieving up to 99% accuracy and capturing complex relationships and interactions. TabNet came second at 96% accuracy, striking a balance between high performance and interpretability that would be especially useful in clinical settings. The study shows how machine learning algorithms can improve heart failure care through early identification, targeted treatment, and enhanced resource allocation in healthcare. The analysis also highlighted significant clinical patterns, such as higher heart failure incidence among older males and elevated cholesterol and blood pressure levels. This cost-efficient approach provides a valuable tool for early intervention, allowing healthcare professionals to identify at-risk patients and improve outcomes. Although the study has some positive findings, it had its limitations. These comprise, in particular, data imbalance, incompleteness in key features, the sensitivity of some models to hyperparameter settings, diminished interpretability in ensemble methods, and limited generalizability across different populations. Future research should focus on expanding datasets and including diverse populations to enhance the model's generalizability.

Keywords: Machine Learning; Convolutional Neural Networks (CNNs); TabNet; Ensemble Methods; Exploratory Data Analysis (EDA)

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Chapter 1 Introduction

1.1 Background and Motivation

Rising heart failure cases have become a major global health issue, and this highlights the importance of early and accurate prediction to facilitate timely treatment and care. It is a challenging problem and requires a precise and efficient solution because of the complexity of medical data and individual variations. This study focuses on developing a robust heart failure prediction (HFP) system using machine learning techniques and attributes derived from the patient's clinical features and medical records.

We have studied multiple ML algorithms, including Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and advanced methods such as Convolutional Neural Networks (CNN) and TabNet. Using algorithms, we have found patients suffering from heart failure with greater precision and dependability than traditional methods such as Naive Bayes.

The strength of our proposed system lies in its ability to accurately identify individuals prone to heart failure by combining different models, thereby benefitting from their strengths. It maximizes medical treatment through better diagnosis and reduces medical costs through precise and efficient allocation of resources. Conducted within a Jupyter Notebook environment using Python, this work is an essential advancement for predictive analytics in healthcare, as it showcases how machine learning can be applied to fundamental medical problems.

1.2 Problem Statement

Cardiovascular diseases, mainly heart failure, are still a significant health burden around the world, with over several million deaths occurring every year. Early diagnosis of heart failure has always been difficult due to the complexity of medical data and conventional statistical methods, even with new developments in diagnostic procedures. Typically, modern prediction algorithms work with minute, homogeneous datasets that suffer from overfitting and a limited generalization capability. Moreover, conventional approaches find it hard to analyze large-scale, multi-dimensional medical datasets with less than satisfactory performance. This work covers these deficiencies and implements machine learning algorithms to build a thorough HFP framework based on diverse patient information. This leads to trustable and interpretable results. By addressing these limitations, this study aims to enhance the quality of early diagnosis, clinical decision-making, and patient outcomes.

1.3 Objectives of this study

This research aims to develop a robust and reliable heart failure prediction system using ML technology. It attempts to help identify early stages of heart failure via patient data (such as medical history, clinical regulations, and lifestyle). The study seeks to predict high-risk patients with potential heart failure, leading to timely medical treatment and personalized treatment strategies to improve patient outcomes. Moreover, it aims to enhance limited medical resources by permitting medical staff to focus on higher-risk cases, reducing monetary wastage, and improving the efficiency of medical treatment. By tackling the global burden of cardiovascular disease through this work, the project reinforces the value of predictive analytics in medicine and plant healthcare.

The specific objectives of this research work are outlined as follows:

(i) Data Integration & Preprocessing:

The primary goal is to integrate various healthcare datasets to form a comprehensive, private, and enhanced merged dataset featuring 23 critical attributes linked to heart failure. The process involves cleaning data, picking relevant features, and applying balancing techniques to ensure high-quality data for the rigorous training of machine learning models. Addressing missing values, outliers, and class imbalance is essential to ensure the dataset is well-suited for building accurate and dependable models.

(ii) Evaluation of Machine Learning Models:

This project assesses and contrasts several trending machine learning models, from the simplest ones, such as Logistic Regression, to more complex models, including Convolutional Neural Networks (CNN) and TabNet. This study identifies the top-performing HFP models by comparing their performance according to metrics including accuracy, precision, recall, and interpretability. CNN attained 99% accuracy, and TabNet balanced high accuracy and clinical interpretability.

(iii) Feature Analysis and Their Relevant Insights:

This study uses statistical analysis and domain expertise to identify the most prominent factors in predicting heart failure. The program examines the key risk factors for heart failure by studying the relationships and interactions between variables such as age, diabetes, chest pain type, and exercise-induced angina. This feature analysis advances model performance and provides healthcare professionals with helpful information.

1.4 Scope and Significance of the Study

This study attempts to develop a HFP system that utilizes a substantial dataset comprising patient demographics, medical records, and clinical measurement data. The core tasks consist of:

- Merging and preparing data sets to eliminate inconsistencies and imbalances.
- Applying statistical and domain-specific approaches for feature selection.
- Assessing multiple machine learning algorithms, including KNN, Logistic Regression, Random Forest, CNN, TabNet, and ensemble methods.
- Establishing the predictive system as a web-based application for real-time applicability in clinical settings.

The specific significances are outlined below:

- **Healthcare Impact:** The study has tools for early detection to bring down the morbidity and mortality rates of heart failure.
- **Technological Advancement:** Using traditional machine learning models and advanced techniques like CNN and TabNet, the project shows how AI is effective in medical diagnosis.
- **Economic Benefits:** This can decrease healthcare costs because of the limited need for unnecessary medical interventions and admissions.
- **Educational Contribution:** The study offers a ground sheet for future research on predictive analytics and its application in the healthcare sector. Additionally, it can contribute to developing personalized medicine and targeted treatments.

Chapter 2 Literature Review

The human heart is a significant organ. It circulates blood throughout the body. If it fails to operate correctly, the brain and other organs will shut down, causing death in minutes. Heart attacks are relatively common. The heart's effect on the circulatory system cannot be overstated. When the heart stops working properly, oxygen cannot reach the body's tissues, leading the circulatory system to collapse. This can result in serious health problems and even death. The terms heart disease and cardiovascular disease have similar meanings. Cardiovascular disease is characterized by restricted or clogged blood arteries, which can cause heart failure, angina, and stroke [1].

The heart is structurally divided into four chambers: two atria (upper chambers) and two ventricles (lower chambers). The valves between these chambers regulate unidirectional blood flow, ensuring efficiency. Functionally, the right side of the heart pumps oxygen-poor blood to the lungs for oxygenation, while the left side circulates oxygen-rich blood to the rest of the body. This continuous and coordinated process is vital for maintaining proper circulation and overall health, as shown in Figure 1.

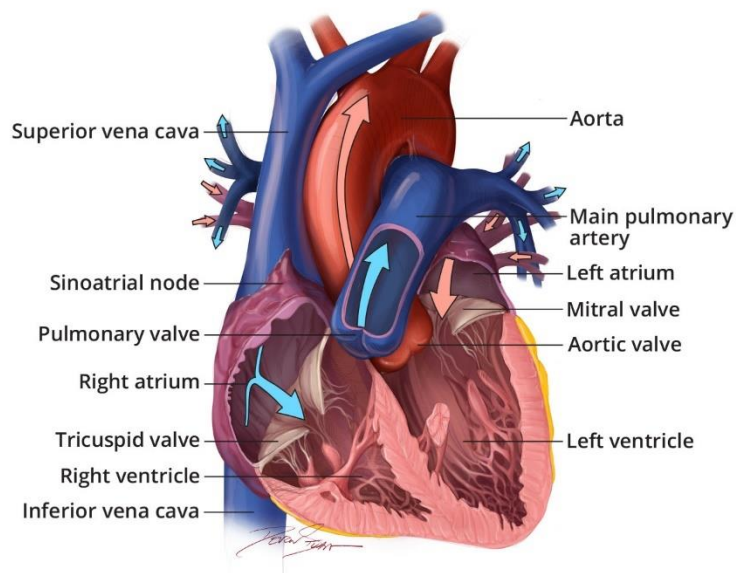
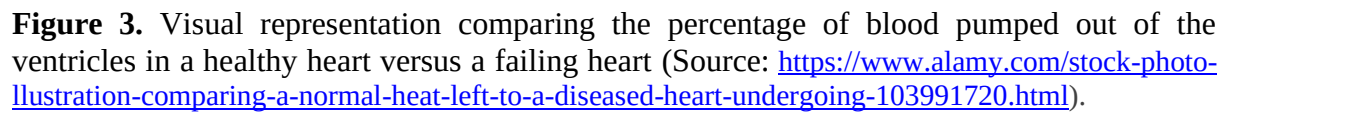
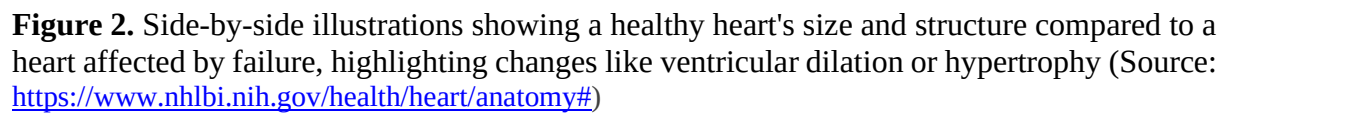


Figure 1. Anatomy of the interior of the heart. This image shows the four chambers of the heart and the direction in which blood flows through the heart (Source: familydoctor.org).

In heart failure, structural changes occur as the heart undergoes remodeling. This can lead to enlarged ventricles (dilation) or thickened heart walls (hypertrophy), which impair its ability to pump blood efficiently. Functionally, the weakened heart muscle struggles to maintain adequate cardiac output,

Heart Failure



A healthy heart ensures efficient blood flow and oxygen supply through coordinated function. In the early stages of heart failure, the heart adapts by thickening its walls (LV hypertrophy) and increasing angiogenesis, often without noticeable symptoms. As the condition worsens, structural changes like ventricular dilation and fibrosis impair pumping efficiency, leading to fatigue, shortness of breath, and fluid retention. This progression is illustrated in Figure 4.

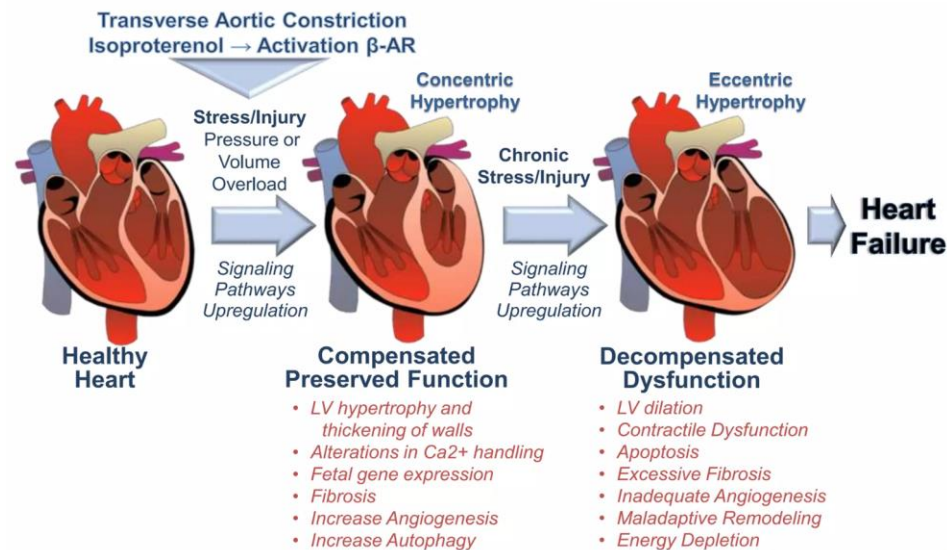


Figure 4. Progression of Heart Failure.

2.1 Literature Review

ML models outperform typical statistical models in predicting mortality and hospitalization risk in patients with cardiovascular disease, according to multiple studies. [2] discovered that an automated machine learning approach outperformed traditional statistical methods in predicting cardiovascular disease risk in a large cohort of UK Biobank participants. [3] found that ML models outperformed traditional methods in forecasting the risk of cardiovascular disease using clinical data. [4] found that ML systems outperformed the GRACE score in accurately predicting the likelihood of death within 30 days following an ST-elevation myocardial infarction.

[5] discovered that merging clinical and cardiac perfusion imaging data into a machine learning model improved prognosis compared to standard approaches. An application for predicting cardiac vulnerability based on primary symptoms was developed using 14 characteristics from Kaggle's UCI Machine Learning Repository. Cardiovascular disease features include age, blood pressure, cholesterol, heart rate, blood sugar, body mass index, old peak, maximum heart rate during exercise, number of significant vessels on fluoroscopy, and serum cholesterol. Categorized attributes include gender, chest pain kind, fasting blood sugar, resting electrocardiography, exercise-induced angina, peak exercise ST

segment slope, thalassemia, and heart disease status. The study compared the accuracy of different machine-learning approaches. ML approaches include LR, NB, SVM with Linear Kernel (LK), RF, and Radial Basis Kernel (RBF) functions. The RF classifier achieved an accuracy rating of 84.81%, surpassing all other ML algorithms tested. The study utilized a 60:40 train-test split for the RF and SVM algorithms [6].

A cardiovascular disease risk prediction study employed data from the UC Irvine repository, comprising 14 features and 303 samples available for public use. The dataset for forecasting cardiovascular illness includes numerical (continuous) parameters, including age, resting blood pressure, cholesterol, maximal heart rate, and ST depression. Categorical features include gender, kind of chest pain, fasting blood sugar, resting ECG result, exercise-induced angina, peak ST segment slope, number of major vessels, thallium stress result, and presence or absence of cardiovascular diseases. An analysis was undertaken to compare MLPro and KNN, two machine learning algorithms. Experiment results show that the MLP model outperforms the KNN model regarding detection accuracy (82.47%) and area under the curve (86.41%). [7] employed an 80:20 train-test split ratio for both machine learning techniques. Databases from the UCI repository, including Statlog, VA Long Beach, Cleveland, Hungary, and Switzerland, were utilized to predict heart disease. The dataset contains features linked to cardiovascular health, including numerical and categorical parameters. Numerical attributes include age, resting blood pressure, serum cholesterol, maximal heart rate, genuine ST depression, and number of colored vessels on fluoroscopy. Categorical features include sex, chest pain type, fasting blood sugar, resting ECG results, exercise-induced angina, peak exercise slope, Thalassemia defect types, and heart disease diagnosis. Performance comparisons were made using the following machine learning approaches: Various algorithms are available, such as RF, NB, DT, KNN, SVM, LR, Gradient Boosting (GB), AdaBoost (AB), and the Bagging Method (BM). The Support Vector Classifier (SVC) outperformed the Decision Tree (DT) with an impressive success rate of 98.31%. According to scientific studies, combining the LASSO feature selection method with additional ML algorithms results in highly accurate and connected features that achieve optimal results across multiple criteria. [8] employed an 80:20 split ratio for training and testing.

The Hungarian and Statlog datasets were used to forecast cardiovascular disease using the Weka algorithm. The Hungarian database, established by the Hungarian Institute of Cardiology in Budapest, comprises 294 instances. The Statlog (heart) dataset contains 304 cases and 76 attributes, although only 14 were used in the tests. The classifiers used are REP Tree, M5P Tree, Random Tree, LR, NB, J48, and JRIP. Performance indicators such as accuracy, MAE, RMSE, and forecast time are computed using the test set. 10-fold cross-validation is a typical performance-measuring approach for predictive ML in practice. Furthermore, the Hungarian database statistics random tree model had a +/- accuracy of 99.81%. A Random Tree model also achieved 100% accuracy on the Statlog database. [9] further validates the effectiveness of predicting cardiovascular diseases using the Random Tree algorithm. Using a heart disease dataset, this paper uses three correlation methods to determine strong relationships between characteristics. The UCI Machine Learning repository's heart disease patient dataset was used. There are 303 examples and 14 attributes in the collection. This study compared the accuracy of KNN,

LR, Gaussian Naive Bayes, Decision Tree, and Random Forest. The study revealed that Gaussian Naive Bayes was better than KNN, LR, RF, and DT in terms of accuracy and area under the receiver operating characteristic curve. Gaussian Naive Bayes outperformed other validation methods, including hold-out validation and 10-fold cross-validation. The study also revealed that the Least Square Linear Regression model performed best in the stratified 10-fold cross-validation with a 70/30 random train test split ratio [10].

An innovative system for predicting cardiovascular diseases has been suggested to employ diverse machine learning techniques and ensemble methods. This study used a combined dataset of five sources: Cleveland, Hungary, Switzerland, VA Long Beach, and Statlog heart disease. The datasets were obtained from UCI's machine learning repository. The merged dataset contains around 1190 cases with 14 unique features, similar to those described in [10]. The classifiers used include DT, RF, KNN, AdaBoost, and Gradient Boosting. The suggested intelligent system accurately predicts cardiovascular sickness, with the Random Forest Boosting Model (RFBM) achieving the highest accuracy rate of 99.05%. The comparison of different classifiers and hybrid techniques showed that RFBM outperformed other models, while K-Nearest Neighbors (KNN) had the lowest accuracy [11]. A heart disease dataset was examined using six different ML algorithms: KNN, DT, SVM, LR, NB, and RF. The Kaggle dataset contains 303 cases with 14 features, including age, resting blood pressure, serum cholesterol level, maximum heart rate, old peak, and fluoroscopy-colored major veins. This dataset includes categorical parameters such as gender, chest pain kind, fasting blood sugar, resting electrocardiogram results, exercise-induced angina, slope, thalassemia level, and target. The accuracy of each classifier was compared using train-test split and k-fold cross-validation methods with different ratios and values. The logistic regression model achieves 82% accuracy with an 80:20 and 75:25 split. SVM achieved 82% accuracy for the 75:25 split, which is comparable. However, when the LR model is combined with the tenfold cross-validation strategy, accuracy rises to 82%. LR and SVM surpass other classifiers [12].

A recent study tested and assessed ML models to predict cardiovascular illness in individuals with heart conditions. The dataset used in the study is easily accessible on Kaggle and was collected by the American Heart Failure Institute. The dataset contains numerical variables such as age, creatinine_phosphokinase, ejection_fraction, platelets, serum_creatinine, serum_sodium, and time, as well as categorical features such as anemia, diabetes, high blood pressure, gender, smoking, and death_event. The list includes various models, including LR, SVM, RF, NB, Gradient Boosting Classifier (GBC), AdaBoost (ABC), CatBoost (CBC), DT, and KNN. The results show that LR outperformed other classifiers with an accuracy of 82.76% in predicting heart disease. ML can improve healthcare outcomes, and the study suggests it could help anticipate cardiac crises [13]. [14] identified the risk factors of coronary heart disease or atherosclerosis using the inbuilt implementation algorithm that employs the same Neural Network techniques and could only accurately predict whether the test patient had the condition. [15] introduced the use of neural networks to diagnose and predict heart disease, blood pressure, and other features. A deep neural network was built using disease-related attributes to produce an output perceptron with 120 hidden layers. This technique ensures accurate results for heart disease when used with the test dataset. The supervised network has been recommended

for diagnosing cardiovascular disorders [16]. When a doctor tests the model with unknown data, the model is used and trained on previously learned data to anticipate the result, determining the model's accuracy. Over the last year, there has been massive growth in data collection and application of machine learning in the healthcare sector, especially in cardiology. These methods are very popular and valuable in the medical field, particularly in cardiology. The abundance of medical information has provided academics with an excellent opportunity to develop and test new methods in this area. Since heart disease is a significant cause of death in underdeveloped countries, it is essential to identify risk factors and early symptoms for study purposes. Data mining and machine learning algorithms can help detect heart disease at an early stage and prevent heart disease.

The research work indicated by [17] is to build an improved machine-learning-based cardiovascular disease (CVD) prediction model for better accuracy than the most widely used Framingham risk score (FRS). Experimental validation of the proposed system, which exploits a quantum neural network to learn and identify patterns of CVD for 689 individuals who displayed CVD symptoms, was compared to the FRS using a second independent validation dataset derived from the Framingham study. The accuracy of the proposed system in predicting CVD risk was found to be 98.57%, which is far better than the FRS accuracy of 19.22% and other existing systems. The authors noted that the proposed method could help doctors predict CVD risk, create better treatment, and enable early diagnosis. Machine learning techniques were applied to predict cardiovascular disease by model development [18]. The data for this was taken from the Cleveland Heart Disease 303 instances of 17 attributes and were sourced from the UCI machine learning repository. The authors utilized several supervised classification algorithms: Naïve Bayes, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN). According to the study results, the KNN model showed the highest accuracy of 90.8%. The prospect of machine learning applications is promising; however, selecting the right model and method remains critical to attaining the best result.

In a study by [19], the key point is that 191 patients with MAFLD underwent blood biochemical analysis and assessment of subclinical atherosclerosis. Machine learning approaches, including multiple logistic regression classifiers, univariate feature ranking, and PCA, were used to construct a model identifying those at most significant CVD risk. The research showed that Heart Failure duration, hypercholesterolemia, and plaque measurements provide the best clinical information for evaluation. By testing 47 potential heart disease patients, the ML method showed 40 (85.11%) correct results, while assessing 144 safe patients, the technique produced a 114 (79.17%) secure result with an AUC output of 0.87. Our analysis shows that ML prediction methods can help physicians find MAFLD patients at high CVD risk using basic medical facts.

A study was used to test the effectiveness of machine learning tools as heart failure disease predictors [20]. The Cleveland Clinic Foundation supplied the study team with their database. The system used multiple learning techniques to make predictions, such as Decision Trees, Logistic regression, Random Forest, Naïve Bayes, and Support Vector Machine. The development process used ten splits of the data to validate performance. The decision tree algorithm delivered 93.19% heart disease prediction success,

surpassing the 92.30% outcome achieved by the SVM model. This research shows how machine learning tools can help doctors predict heart failure patients. Our study shows that the decision tree algorithm could become a strong focus for heart disease prediction research.

The authors studied different ML systems to establish which one selects cardiovascular disease features most efficiently [21]. To begin the analysis, the study team chose all three fundamental methods of feature selection: filter, wrapper, and embedding. The researchers collected features from all three techniques before they selected only the best-performing set using boolean alignment. The sequence comprised two steps to find feature subsets. Several models, including random forest, support vector classifier, k-nearest neighbors, naive Bayes, and XGBoost, were considered to justify the comparative accuracy and identify the best predictive analytics. The artificial neural network (ANN) was used as a standard for comparison with all features. The findings demonstrated that the most accurate prediction results for cardiovascular illness were provided by the XGBoost classifier coupled with the wrapper technique. XGBoost delivered an accuracy of 73.74%, followed by SVC with 73.18% and ANN with 73.20%.

2.2 Limitations of the previous studies

The primary drawback of the prior research is its limited dataset, resulting in a high risk of overfitting. The models developed may not be appropriate for large datasets. In contrast, we utilized a cardiovascular disease dataset of 70,000 patients and 11 features, reducing the chance of overfitting. Cardiovascular diseases are prevalent these days. They describe a range of conditions that could affect our hearts. According to the World Health Organization, 17.9 million people die each year as a result of CVDs [22]. Our study can identify which persons will likely be diagnosed with heart disease based on their medical history. It recognizes who has any indications of heart illness, such as chest pain or high blood pressure. It can help identify diseases with fewer medical tests and more effective therapies, allowing them to be treated appropriately.

This project aims to check whether the patient is likely to be diagnosed with any cardiovascular heart disease based on their medical attributes such as gender, age, chest pain, and fasting sugar level. A dataset with the patient's medical history and attributes is selected from the UCI repository. Using this dataset, we predict whether the patient can have a heart disease. To predict this, we use 14 medical attributes of a patient and classify him if the patient is likely to have a heart disease. These medical attributes are trained under three algorithms: Logistic regression, KNN, and Random Forest Classifier. The most efficient of these algorithms is KNN, which gives us an accuracy of 88.52%. Moreover, we classify patients at risk of getting heart disease or not in this project. [23]

Chapter 3 Methods and Materials

3.1 Study Design

This project included three datasets and identified 23 significant factors related to heart failure. We used exploratory data analysis (EDA) to find patterns and relationships, including the higher risk of heart failure in older adults and males—the frequency of typical angina in heart disease patients. We cleaned the dataset by standardizing and eliminating 30 indices, resulting in 368 instances. The dataset was divided into training and testing sections in an 80/20 ratio. We used several machine learning models, including Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors, Random Forest, Bagging Decision Tree, Voting Classifier, Tabnet, and CNN models. The CNN classification model stands out by delivering 99 percent accuracy. We have applied established steps to prepare data and select features before validating prediction outcomes. The details of the project are given in Figure 1.

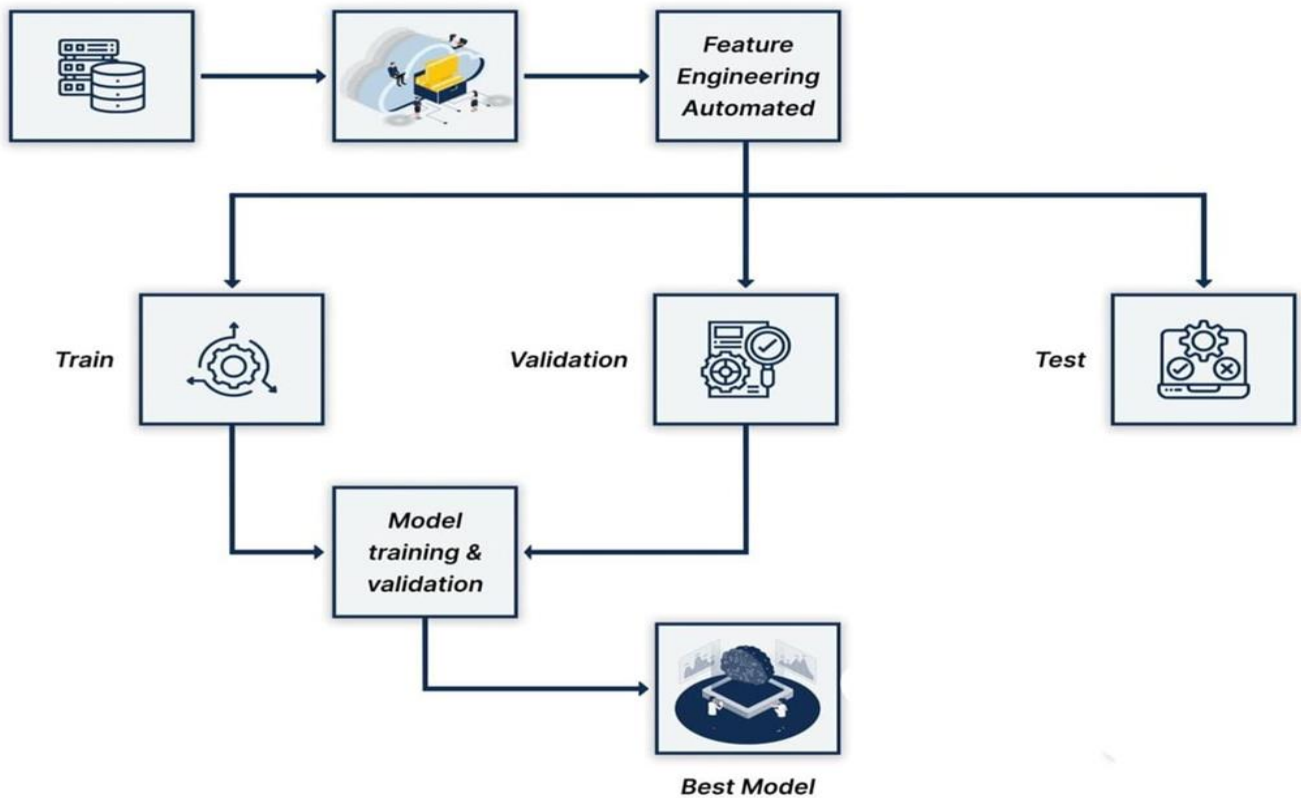


Figure 5. Proposed System Design Diagram of HFP using ML.

We balanced the dataset after EDA to prevent models from predicting more often in favor of one class over the other. Standardization normalized the dataset for our applied machine-learning techniques to work with.

Before starting the experiment, we separated our fixed preparation dataset into 80% training data and 20% test data blocks. We split our available data to train multiple models until we reached our desired ratio for unbiased model performance evaluation. We have tested how well Logistic Regression, Gaussian Naive Bayes, KNN, Random Forest, and multiple additional machine learning methods performed. The model results reached their highest efficiency through custom parameter settings. Our experiments included changes to neighbors and trees' KNN and Random Forest parameter settings, respectively. We measured how accurately each model predicted by calculating model accuracy while validating results with cross-validation. Logistic regression proved the best prediction model, with Bagging Decision Trees and Voting Classifiers close behind among the machine learning methods. Out of all tested models, the CNN showed the highest achievement rate. We judged each results collection through their accuracy with prediction tests. Our detailed work created a dependable prediction model that will serve as the foundation for future improvements and real-world implementation.

3.2 Dataset Collection

Our research includes three separate healthcare databases that deliver key heart failure data. Each dataset included basic patient data, complete health records, and clinical results. Our research successfully combined these datasets to produce an open and reliable data source that capitalized on each resource's unique features and depth of data—merging required matching equivalent traits, resolving inconsistencies, and ensuring data consistency.

First dataset source: Davide Chicco, Giuseppe Jurman: "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone." BMC Medical Informatics and Decision Making 20, 16 (2020).

Second dataset source: This dataset has been collected from a hospital in Pakistan located in Faisalabad named the Institute of Cardiology.

Third Dataset source:

Reference: International application of a new probability algorithm for diagnosing coronary artery disease.

Dataset Description

This data set consists of 23 significant attributes that are key in building an effective predictive model for heart failure. The features are:

- **Age:** The age of the individual, measured in years. Age is a critical factor as the risk of both heart disease and heart failure increases significantly with age.

- **Sex:** The biological sex of the individual (male or female). Males generally have a higher risk of developing these conditions compared to females.
- **Family History:** Whether there is a history of heart disease or Heart Failure in the individual's family. Genetic predisposition plays a crucial role in these conditions.
- **Hyperlipidemia:** High blood lipids increase the chance of heart disease and cardiovascular problems.
- **Heart Failure:** The medical report shows if the patient received a heart failure diagnosis. Heart failure appears either as a present or absent condition in our data.
- **Allergies:** The list of recognized allergies that impact overall wellness and medicine selection.
- **Depression:** Depressive symptoms show up through official healthcare records and patient responses to questionnaires. Having depression makes health worse and creates more danger of developing heart disease.
- **Hypertension (HTN):** Having blood pressure readings that are too high puts you at risk for heart problems.
- **High Blood Pressure:** Recent blood pressure recordings, including its two main numbers, systolic and diastolic values.
- **Life Style:** The assessment of online lifestyle factors checks the person's physical activities, eating habits, smoking patterns, and drinking habits.
- **Locality:** People in particular areas must deal with environmental risks and healthcare service limitations.
- **Chest Pain Type:** Angina types fall into four categories: Typical Angina, Atypical Angina, Non-Anginal Pain, and Asymptomatic. Patients who suffer from typical angina nearly always show signs of coronary artery disease.
- **Blood Pressure (BP):** Physical tests help doctors understand the heart's performance by measuring blood pressure numbers.
- **Cholesterol:** Testing shows blood cholesterol numbers for total cholesterol plus HDL and LDL counts.

- **Diabetes:** A positive diabetes test shows if someone has had diabetes before because the condition weakens blood vessels and harms heart health.
- **Fasting Blood Sugar (FBS) over 120:** This shows when your fasting blood sugar level hits 120 mg/dL, which points to Heart Failure risk or when you are on the edge of developing that condition.
- **Electrocardiogram (EKG) Results:** An EKG test records the heart's electrical activity and can indicate abnormalities.
- **Maximum Heart Rate (Max HR):** The highest heart rate achieved during physical exertion, which can reflect cardiovascular fitness and health.
- **Exercise Angina:** Whether chest pain develops during physical activity.
- **ST Depression:** An electrocardiogram's downward movement of the ST segment indicates heart tissue problems from insufficient blood.
- **Slope of ST:** The ST segment rise or decline pattern in cardiac monitoring affects patient outcomes in specific ways.
- **Number of Vessels Fluro:** A complete count of arteries was displayed on a fluoroscopy, and severe stenosis was observed.
- **Thallium:** During a Thallium exam, doctors evaluate heart muscle blood flow patterns to identify areas receiving insufficient blood.

3.2.1 Feature Extraction Techniques

Heart failure analysis depends heavily on feature extraction techniques since they reveal important picture information to create accurate patterns. These methods span different ways to obtain data by either developing specific visual patterns manually or teaching the system to find essential patterns. Experts use their knowledge to develop handcrafted features that describe specific medical image attributes, including texture patterns and brightness strength. Human-made design elements bring easy understanding to medical imaging review but cannot detect small image changes effectively. Deep learning through CNN exercises feature extraction while using raw pixel information to build hierarchical representations. CNNs better detect intricate medical image patterns by building improved feature versions across multiple abstraction stages, which help the classifier work better. Learning representations prove more effective than standard handcrafted features at identifying subtle changes in medical images of diseased tissue.

3.2.2 Feature Selection

Our model performance improved by choosing 23 top features from all available data attributes. The selection of features followed a structured series of actions (Table 1). Our categorical and continuous features analysis used chi-square tests with *t*-tests to show which elements were important. Our analysis of feature correlation used matrices to find which features most closely matched our target outcomes (heart disease and heart failure). Significant feature overlap was checked to eliminate unnecessary measurements from our analysis. We cleaned the data by removing the rows with NaN values. Also, we have checked the duplicate values and removed those entries (Table 1).

Table 1. Selected Features, with count number and data type.

Feature	Non-null Count	Data Type
Age	368	int64
Sex	270	float64
Chest pain type	270	float64
BP	368	float64
Cholesterol	270	float64
FBS over 120	270	float64
EKG results	270	float64
Max HR	270	float64
Exercise angina	270	float64
ST depression	270	float64
Slope of ST	270	float64
Number of vessels fluro	270	float64
Thallium	270	float64
Heart Disease	270	object
Family History	368	object
Life Style	368	object
Locality	368	object
Hyperlipidemia	368	object
Diabetes	368	int64
Allergies	368	object
Depression	368	object
HTN	368	object

Note:

Inputs after balancing data: **338**

Targets after balancing data: **338**

Domain Expertise: First, we consulted with healthcare professionals and reviewed medical literature to identify features that are known to influence heart disease and heart failure.

Statistical Analysis: We used chi-square tests for categorical data and t-tests for continuous features to establish the relevance and significance of each feature.

Correlation Analysis: We used correlation matrices to select features with the highest correlation to target variables (Heart failure). Features with significant multicollinearity were carefully evaluated to avoid redundancy.

3.3 Machine Learning Methods with Their Description

Model Descriptions and Mathematical Equations

This study evaluated several machine learning models to predict heart failure. The models used were Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors (KNN), Random Forest, Bagging Decision Trees, and Voting Classifier. Details about each model and their mathematical formulations are given below:

3.3.1 Logistic Regression

Logistic regression is a statistical method used to solve binary classification problems. It uses the logistic (sigmoid) function to model the connection between the independent and dependent variables and estimate the likelihood that a given input belongs to a specific class. This function converts the linear combination of input features into a probability value between 0 and 1. Logistic Regression is highly interpretable since the coefficients show how each feature contributes to the likelihood of the desired outcome. For this study, Logistic Regression was trained by dividing the dataset into an 80-20 train-test split, with preprocessing techniques such as feature standardization used to scale input variables. This model performed well for linearly separable features. The equation of the logistic regression model is given below:

$$P(Y = 1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n)}} \quad (1)$$

where, $P(Y = 1|X)$ is the probability that the outcome Y is 1 given the predictor variables X , β_0 is the intercept term, $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients for the predictor variables $X_1, X_2, \dots, X_k$ and e is the base of the natural logarithm.

3.3.2 Gaussian Naive Bayes

Gaussian Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem that assumes all features are conditionally independent of the class label. Furthermore, it assumes that the continuous features have a Gaussian distribution. This simplicity makes Gaussian Naive Bayes computationally efficient and appropriate for small to medium-sized datasets. Despite its precise assumptions, it works effectively when the independence assumption is approximately valid or severe correlations are absent. Despite its simplicity, it attained an accuracy, which served as a baseline for comparison with more complex models. The equation of gaussian naïve bayes model is given below:

$$P(C_k|X) = \frac{P(C_k) \prod_{i=1}^n P(X_i|C_k)}{P(X)} \quad (2)$$

3.3.3 K-Nearest Neighbors (KNN)

KNN is a non-parametric, instance-based learning technique used for classification and regression. The model does not learn a precise decision boundary during training, which makes it memory-intensive because it must store the entire dataset. KNN performs well with small datasets and unambiguous class separations but can be computationally expensive with larger datasets. The choice of k and feature scaling significantly impacts performance. The model was trained using cross-validation to maximize the hyperparameter k (number of neighbors). The dataset was standardized to guarantee that all features received equal weightage. KNN obtained a moderate accuracy. However, its performance was limited by high processing demands and sensitivity to the distance metric. The equation of the KNN model is given below:

$$y = \text{mode}\{y_i | i \in N_k\} \quad (3)$$

3.3.4 Random Forest

During training, Random Forest creates multiple decision trees that combine at runtime to solve classification or regression problems. Reducing overfitting helps models predict better across new situations. The algorithm of Random Forest resists noisy data and shows how different features work together through complex non-linear relations. The method provides valuable data about which features matter most so we can use it to find and select significant features. It requires high resource usage for big datasets, whether they contain multiple trees. The model learned from subsets of data

and features through bootstrapping, which created different trees that prevented data from overfitting. The model showcases that it can analyze complex connections between different attributes. The equation of the Random Forest model is given below:

$$MSE = \frac{1}{N} \sum_{i=1}^N (f_i - y_i)^2 \quad (4)$$

3.3.5 Bagging Decision Trees

Using a sampling method called Bagging (Bootstrap Aggregating), multiple decision trees learn from different parts of the training data. These distinct trees combine their predictions to create a final model output. By combining multiple decision trees, Bagging reduces errors and enhances stable predictive performance. The method produces more stable predictions than one decision tree because it deals better with datasets that fluctuate significantly. Although Bagging performs like Random Forest, it fails to handle dependencies in the data because it does not select random features during splits. We used identical data splitting and pre-processing steps in this model. Our model shows better results in fighting overfitting than just using individual decision trees. The equation of the Bagging Decision Trees model is given below:

$$\hat{f}(x) = \frac{1}{M} \sum_{m=1}^M f_m(x) \quad (5)$$

3.3.6 Voting Classifier

Voting Classifier combines predictions from multiple machine learning models to deliver better results. The Voting Classifier performs majority voting to combine predictions in classification setups yet runs average calculations for regression work. The approach selects different base models to enhance stability based on their respective capabilities. Users can employ linear and nonlinear models with a Voting Classifier to handle their unique datasets. Although easy to set up, the approach depends on having well-performing foundation models. The model uses Logistic Regression and Random Forest alongside GNB by taking advantage of each class's strengths. The training model worked with prepared data, showing its digital universality. The equation of the Voting Classifier model is given below:

$$\hat{f}(x) = \text{mode}\{f_1(x), f_2(x), \dots, f_M(x)\} \quad (6)$$

3.3.7 TabNet

TabNet is a specialized deep-learning structure built entirely for processing tabular information. The method applies moving attention systems to locate and study relevant information for every step in the decision-making process. The model proves its strength when handling data with complex connections and multiple kinds of features. Classic neural networks often need extended preparation work for tabular data, but TabNet handles this data type efficiently. Its outcome depends on modifying system settings and needs substantial processing power. We split the dataset for analysis and prepared it through TabNet training on a learning rate scheduler and a regularization scheme. It achieved 96% accuracy through a trade-off between performance and explainability features. The equation of Tabnet model is given below:

$$Z_t = f_{shared}(x) + f_{independent,t}(x) \quad (7)$$

3.3.8 CNN (Convolutional Neural Network)

A deep learning network called CNN was created to process images, yet it works well with tabular data. They apply convolutional layers over the input data to identify and extract significant patterns before feeding them to their model. CNNs are highly adept at collecting local and global trends due to their hierarchical feature learning capacity. Their deep design enables them to model complicated, non-linear interactions. However, their interpretability is limited compared to simpler models, making them less appropriate for transparent decision-making applications. This study used the preprocessed dataset to train CNNs using backpropagation and optimization approaches such as the Adam optimizer. The model reached the most remarkable accuracy, successfully capturing complex feature relationships and patterns. The equation of the CNN model is given below:

$$Y_{i,j} = \sum_{m=1}^{k_h} \sum_{n=1}^{k_w} \sum_{c=1}^C X_{i+m-1,j+n-1,c} \cdot K_{m,n,c} \quad (8)$$

3.4 Tools, Equipment, and Software Used

IntelliJ IDEA: IntelliJ IDEA is a Java IDE for developing software. JetBrains (previously IntelliJ) developed it, and it is available in an Apache 2 Licensed community edition and a proprietary commercial edition. Both can be utilized for commercial purposes.

PyCharm: PyCharm is an integrated development environment (IDE) for computer programming, specifically Python. It was created by the Czech company JetBrains. It includes code analysis, a graphical debugger, an integrated unit tester, version control system (VCS) integration, and Django web

development support. PyCharm has intelligent code completion, code inspections, on-the-fly error highlighting and rapid fixes, automated code refactoring, and extensive navigation capabilities: Python, JavaScript, CoffeeScript, TypeScript, CSS, popular template languages, and more. Use language-aware code completion, error detection, and on-the-fly code fixes.

Smart search allows us to navigate to any class, file, symbol, IDE action, or tool window. It only takes one click to get to the declaration, super method, test, usages, implementations, and other sections. We used PyCharm to run the server-side code in Python and process the entire system. We can manage our functionalities with this software because it has a graphical user interface (GUI).

Anaconda: Anaconda is a free, open-source Python and R programming language that facilitates package management and deployment. The package management system Conda manages package versions. Over 15 million users utilize the Anaconda distribution, which includes over 1500 popular data science packages for Windows, Linux, and MacOS. The Anaconda distribution includes more than 1,500 Conda packages and virtual environment managers. It also contains Anaconda Navigator, a graphical alternative to the command line interface (CLI). Conda analyzes our current environment, everything we have installed, and any version limitations we specify (e.g., you only want TensorFlow >= 2.0) and figures out how to install compatible dependencies.

Jupyter Notebook: Jupyter Notebook is an open-source interactive development environment for data analysis, machine learning, and research. It was chosen for this project because it enables the seamless integration of code, written explanations, visualizations, and outputs into a single document. This feature allows for greater cooperation, information, and documentation of the project workflow. Jupyter Notebook supports various programming languages. It is adaptable and ideal for integrating Python libraries for data preprocessing, feature analysis, and machine learning model creation. Users can enter code and test their work immediately because Jupyter Notebook shows output directly in graphs, tables, and results. Our data exploration and model refinement processes use this feature very effectively. The study chose Jupyter Notebook for its easy setup and support of TensorFlow, Pandas, and Matplotlib, which works alongside the ability to make duplicate research findings.

TensorFlow and Keras: The project selected TensorFlow and Keras because these open-source frameworks perfectly suit all machine learning model development needs. The machine learning system runs deep learning programs with TensorFlow power. At the same time, Keras gives users an easy way to create models through its user-friendly interface. Our chosen tools uphold neural networks well and support specialized architecture such as this project's CNN. The dynamic functionality of TensorFlow makes it easy to work on massive datasets and advanced mathematical processes. However, Keras helps developers build programs more quickly because it offers easy-to-use components to test and fix issues. Their combined services deliver smooth GPU-powered training capabilities alongside rich documentation to help developers create effective HFP models at scale.

Hardware and/or Software Implementation:

- **Hardware implementation:**

Through hardware insertion, users can create efficient processing systems to run calculations. Our system combined the NVIDIA GPU 3050 with other hardware blocks, and this GPU served as the fastest training resource. The CPU and memory worked with other hardware parts, including the CPU and storage devices, for typical system performance during setup.

- **Software Implementation:**

We focused on developing and perfecting all modules within our software. We trained deep-learning models through TensorFlow and Keras while using simulations to tune hyperparameters and verify model stability. Jupyter Notebook made scripting and model testing easier to help developers create and facilitate iterative model development.

3.5 Data Analysis and Preprocessing

Our research used advanced data processing methods to build stable machine-learning models that predict heart failure outcomes. A single data source merged three healthcare records with patient demographics, health data, and lifestyle and medical measurement reports. The combined dataset expanded the available feature selection, enhancing model prediction accuracy.

The starting point of our data analysis was exploratory data analysis (EDA), which sought to reveal major data trends, linkages, and potential errors in the available information. Our analysis evaluated numeric statistics from various data viewing methods like statistical tables plus graphs of distribution shape. Our research reveals that heart failure risk rises with both advanced age and high cholesterol levels in patients who report typical angina symptoms. Through EDA, the team found that men develop higher heart failure risk than women, which matches existing medical knowledge. The target variable relationship became clear through histograms and scatter plots, which helped us choose features of better quality.

Our dataset preparation process included multiple processing actions before training our ML model. Our data preprocessing methods dealt with incomplete data values in ways that preserved the original data structure without sacrificing important details. We converted category data values into numbers through one-hot encoding. The target variable's imbalance problems were reduced using over-sampling and under-sampling methods, leading to better model results. We have standardized and normalized

numerical features to make them compatible with all analysis techniques at the correct scale. Careful analysis of z-score and boxplot-detected outliers ensured their elimination to improve model accuracy.

The dataset was partitioned into training and testing subsets to enable fair model evaluation in an 80:20 split. Our system used cross-validation methods to set the best model parameters and prevent excess training data effects. Our domain experts and statistical tests generated 23 enhanced features through feature engineering. Reviewing and preparing our data effectively set us up for developing strong machine-learning models. Our process produced a filtered dataset that realistically showed increased precision in heart failure risk predictions from important model data.

Chapter 4 Results

4.1 Descriptive Analysis

Exploratory Data Analysis

A multi-channel Exploratory Data Analysis (EDA) was run to visualize the different features in the dataset, identify relationships, and extract meaningful patterns. It was a decisive moment in our model and feature selection decision-making process.

- **Population Analysis Reveals Aging Trend/Analyzing the Heart Failure Age Histogram**

The age histogram indicates that the incidence of heart failure is strongly correlated with age and that most of the incidence falls on the right tail of the data. Heart failure most commonly occurs in these patients in the age group 55–60, who have an average age of 54 (Figure 2). That trend indicates that the older you are, the greater the risk for heart failure. These factors include age-related physiological changes, pre-existing health conditions like high blood pressure and diabetes, and lifestyle attributes such as smoking and bad diet.

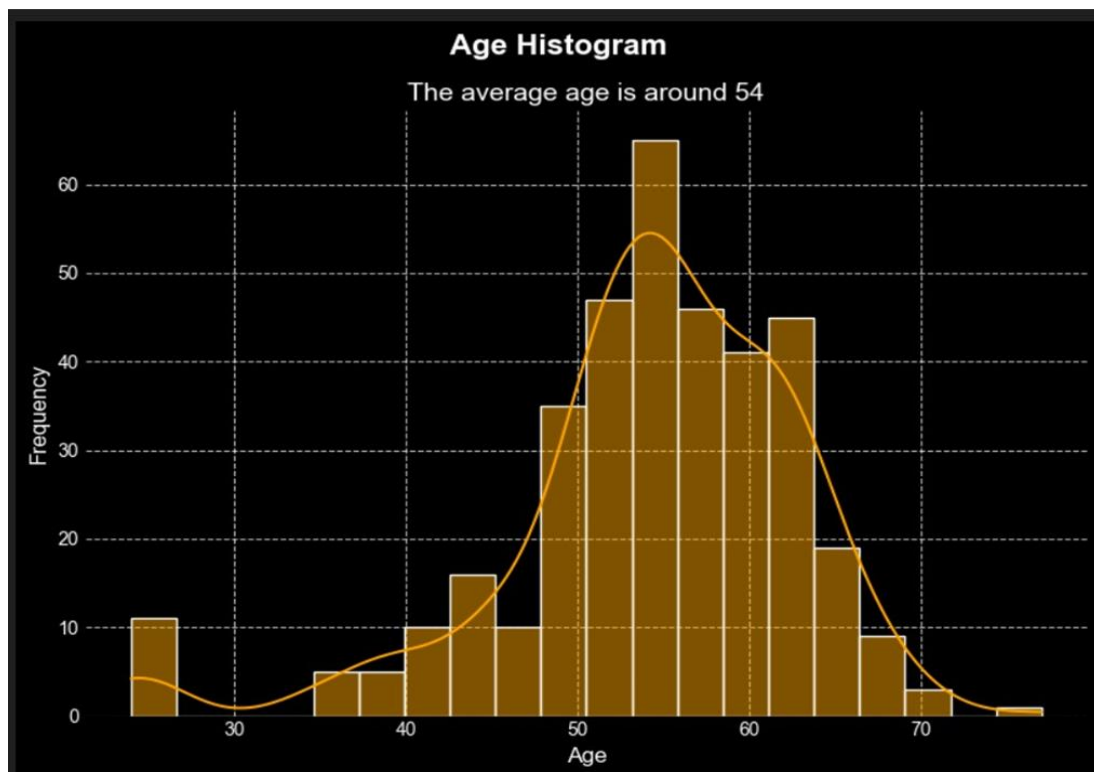


Figure 6. Histogram of Age.

Distribution of Heart Failure Patients. Here, the X-axis is the age group, and the Y-axis shows the frequency, which is the number of patients with heart failure in that age group.

- **Analyzing the Heart Failure Gender Distribution**

As we can see from the gender distribution data below, the study has a male population of roughly 70% vs 30% female (Figure 3). This skewness shows that heart failures are often found in the male population. We can see a vast difference in the count plot, showing nearly 175 males, whereas 85 females were affected by heart failure (Figure 3).

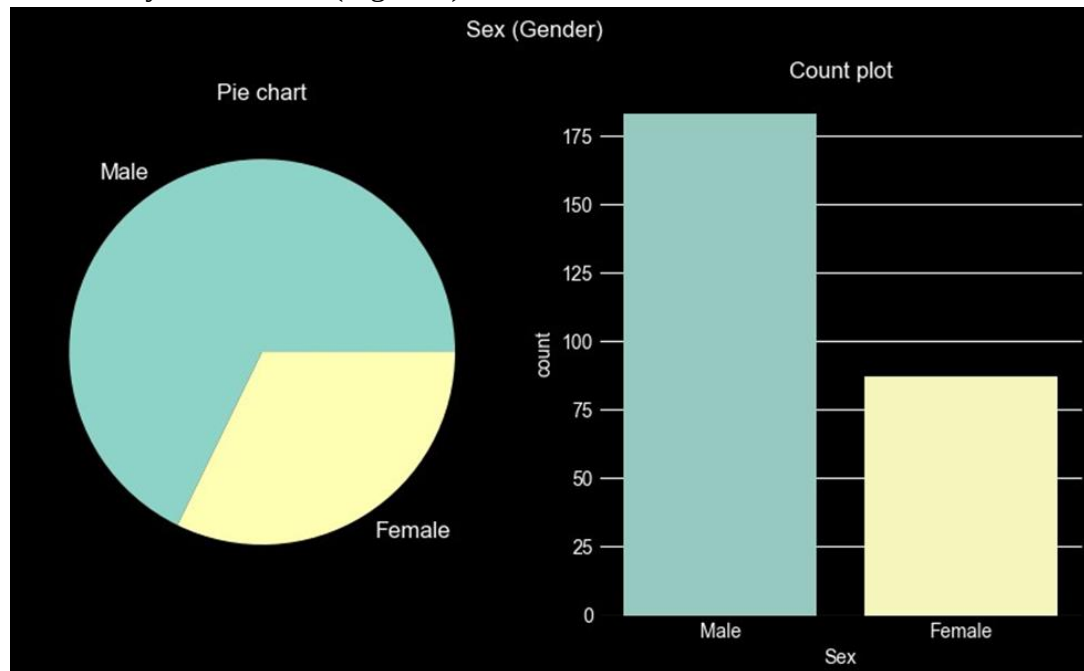


Figure 7. EDA of Gender.

Gender Distribution of Heart Failure Patients. The pie chart shows the distribution of males and females in percentage terms. In contrast, the count plot shows the number of patients, which reveals that males are the majority in our data.

- **Chest Pain Type Distribution in Heart Failure Patients**

The pie chart and bar plot show various chest pain types distribution for heart failure patients. The majority of these patients (47.8%) had typical angina, while 29.3% had atypical angina. 15.6% of patients encounter non-anginal pain, and 7.4% remain asymptomatic (Figure 4). What the findings tell us is that chest pain is common among people with heart failure, and it is typical angina that dominates. It creates a picture of possible subtypes of chest pain recognition that can be vital in the early indication or critical management of heart disease.

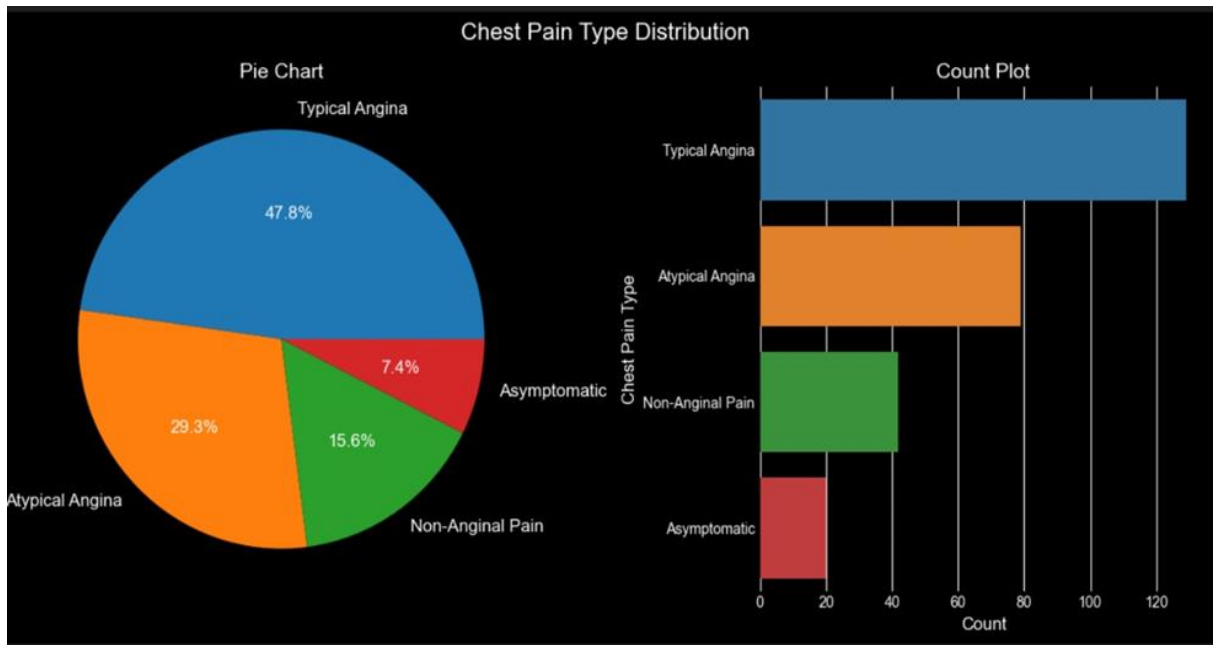


Figure 8. EDA of Chest Pain Type.

Distribution of Chest Pain Types Among Heart Failure Patients. The pie chart highlights the percentage of patients with each chest pain type. On the other hand, the bar plot represents the number of patients in each category.

- **Resting Blood Pressure Analysis in Heart Failure Cases**

Rest blood pressure (BP) data analysis demonstrates extensive spread among patients. The interquartile range of BP is between the first and third quartiles of BP, with some outliers falling below 80 mmHg and above 160 mmHg. The histogram shows that most patients fall within the range of 100 to 120 mmHg (Figure 5). A considerable number of patients also present with BP values that threshold up to 140 mmHg, which may signal an increasing trend in resting blood pressure in the general population.

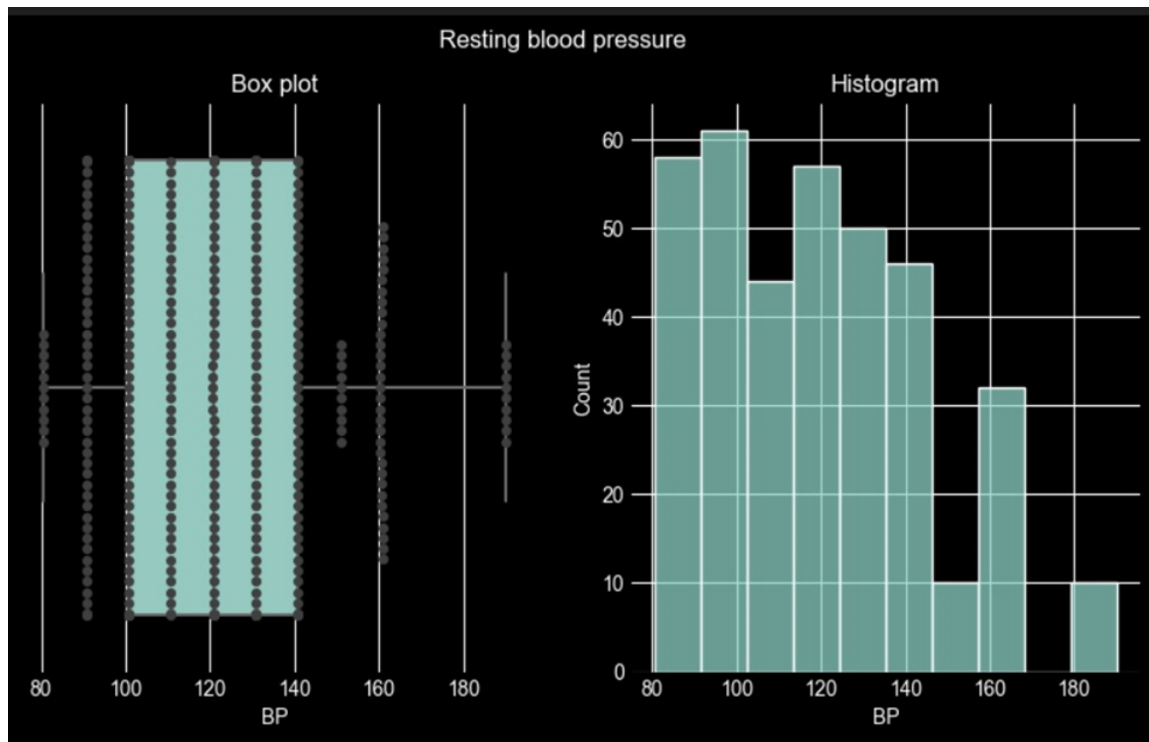


Figure 9. EDA of resting blood pressure.

Distribution of Heart Failure Patients. BP values spread on the left box plot (the left image) with outliers, and their frequency distribution across all participants (the right image) is shown with a peak at ~100-120 mmHg.

- **Cholesterol Distribution and Variability in Heart Failure Patients**

We see the same trend in the above box plot and histogram of cholesterol levels among heart failure patients. The left box plot shows the interquartile range and that most cholesterol values lie between 200 and 300 mg/dL (Figure 6). A few outliers exist, with individual values above 500 mg/dL. The histogram on the right shows that the most common cholesterol level is around 250 mg/dL and that there is a rightward skew (Figure 6). These findings indicate that high cholesterol levels are typical in heart failure and may underlie the cardiovascular risk in these patients.

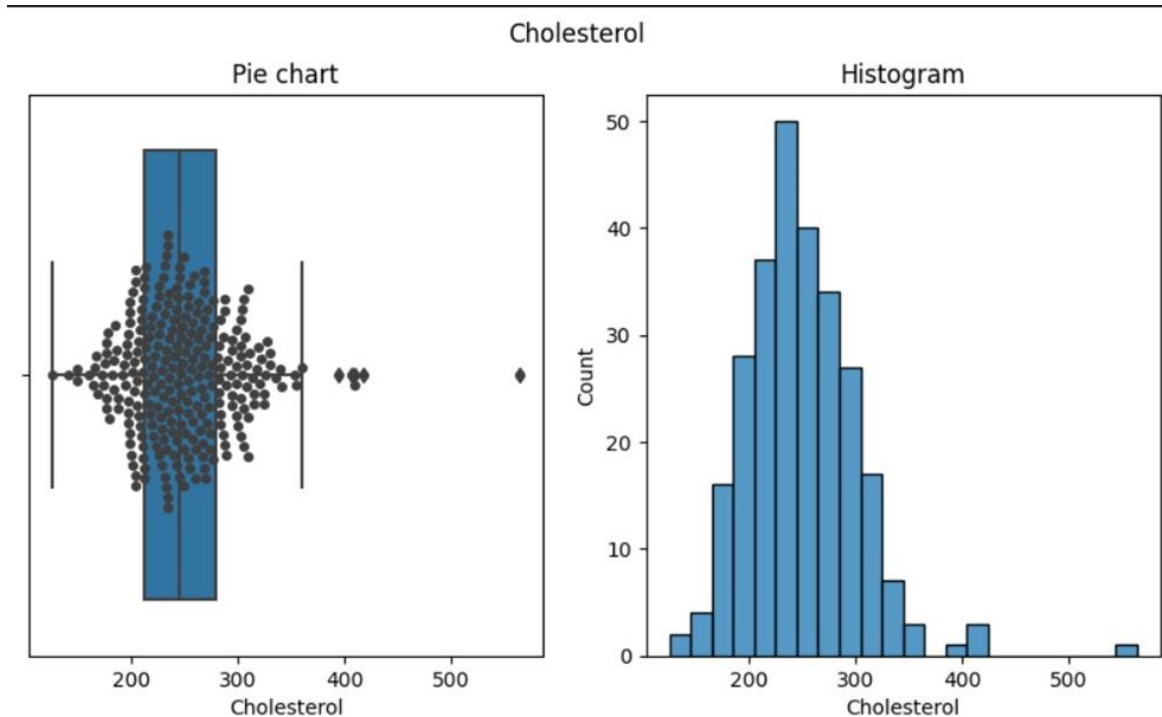


Figure 10. EDA of Cholesterol.

Cholesterol Levels in Heart Failure Patients. The box plot illustrates the distribution and any outliers in the cholesterol data, and the histogram shows the distribution of cholesterol levels within the patient population.

- **Fasting Blood Sugar (FBS) Levels Over 120 in Heart Failure Patients**

The pie chart and count plot show the categorization of heart failure patients according to FBS (fasting blood sugar) level. Most patients (about 85%) have FBS levels below 120 mg/dL, and a smaller subset (roughly 15%) has FBS levels above 120 mg/dL (Figure 7). This indicates that most heart failure patients do not have elevated blood sugar. Still, a significant number do, suggesting that high fasting blood sugar may be a risk factor for heart failure. Below are some contributing factors that cause FBS to be higher: Existing conditions such as diabetes and poor dietary choices.

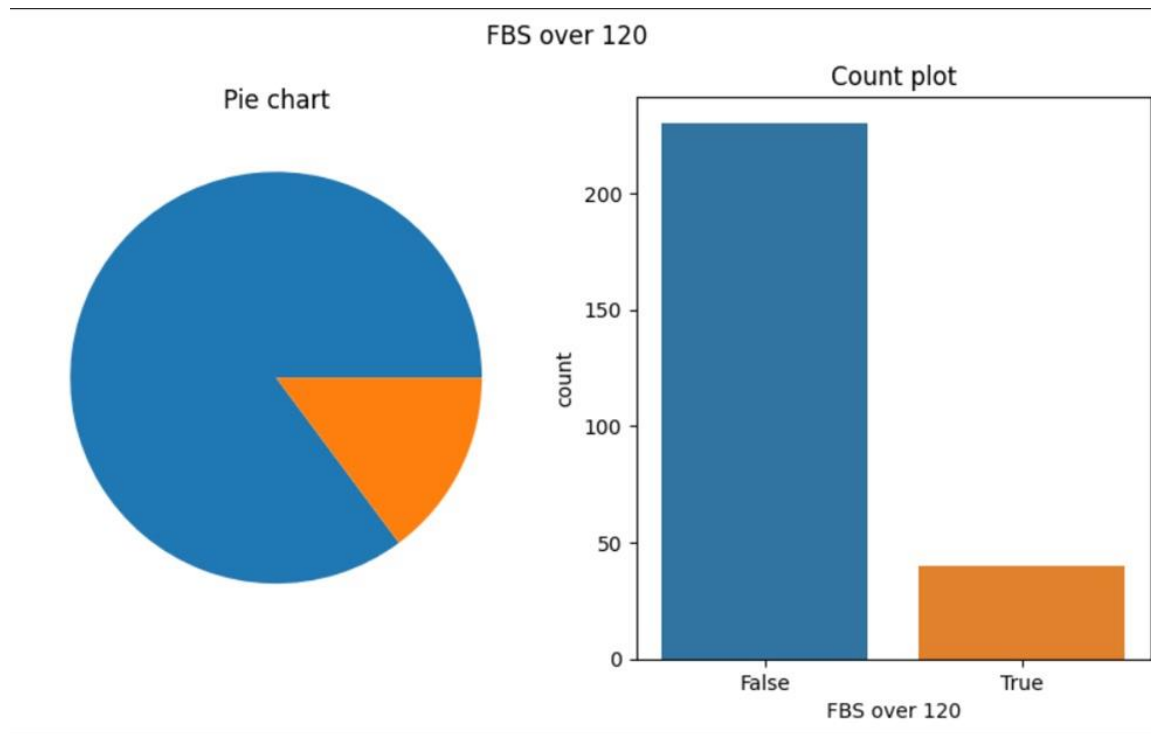


Figure 11. EDA of FBS Over 120.

Distribution of Fasting Blood Sugar Levels in Heart Failure Patients. The pie chart shows the percentage of patients with an FBS level greater than 120 mg/dL in their blood, while the count plot gives a clear number of patients on each side of the threshold.

- **Electrocardiogram (EKG) Results Distribution in Heart Failure Patients**

The distribution of heart failure patients by EKG result consists of a pie chart and count plot on the above figure. Data shows that most patients have their EKG as usual (0) (around 47%). Nearly half (~50%) have abnormal EKG (2) results (Figure 8). A very low percentage (around 3%) of patients have borderline (or dubious) results (marked with 1). This is not a common finding – previous studies have shown that nearly half of the patients admitted with heart failure have an abnormal EKG, likely because heart failure is often due to pre-existing heart conditions, which have their EKG-identified electrical abnormalities.

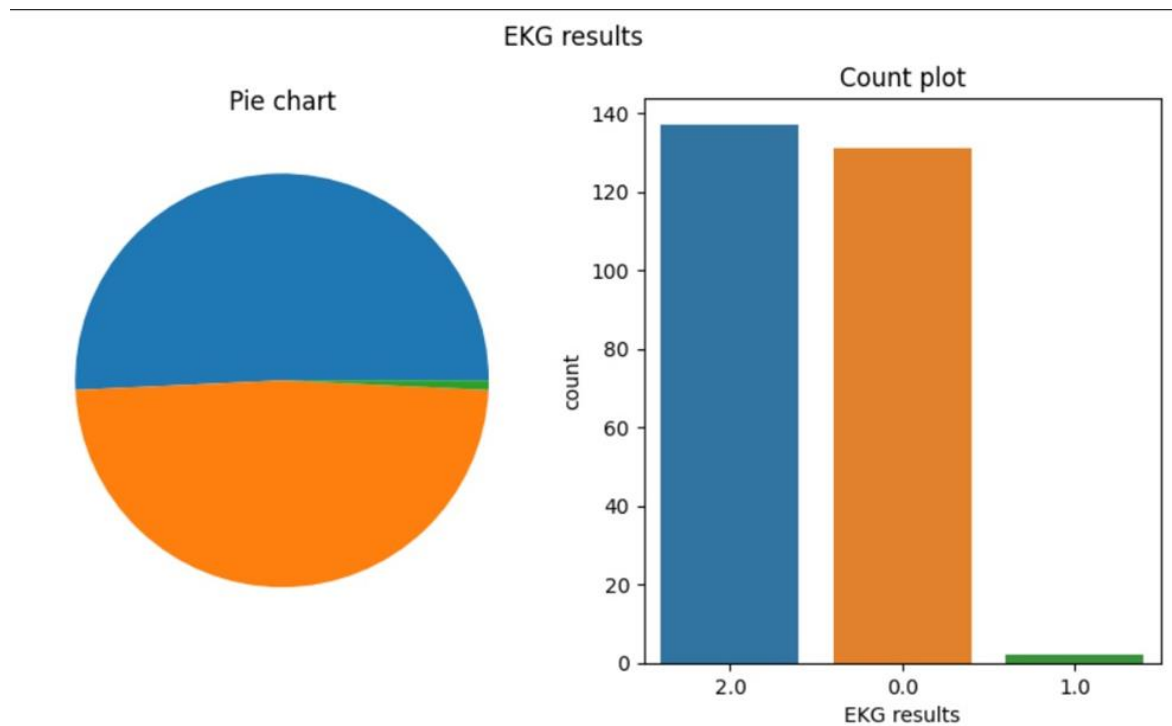


Figure 12. EDA of EKG results.

Distribution of Electrocardiogram (EKG) Results in Heart Failure Patients. This pie chart allows us to visualize the categories of EKG results, and the count plot shows us how many patients fall in that respective result group.

- **Analyzing Maximum Heart Rate in Heart Failure Patients**

Data visualization represents the distribution of heart failure patients' maximum heart rates (Max HR). The box plot shows that most patients have a Max HR between 130-170 bpm, with a median of around 150 bpm (Figure 9). The histogram shows that the maximum heart rates are frequently between 150 and 160 bpm, indicating that patients with heart failure reported higher values of maximum heart rates (Figure 9). However, a few patients have significantly lower or higher Max HR, representing outliers in the data. The overall trend indicates that patients with heart failure tend to experience increased heart rates, which could be linked to compensatory mechanisms in response to poor cardiac function.

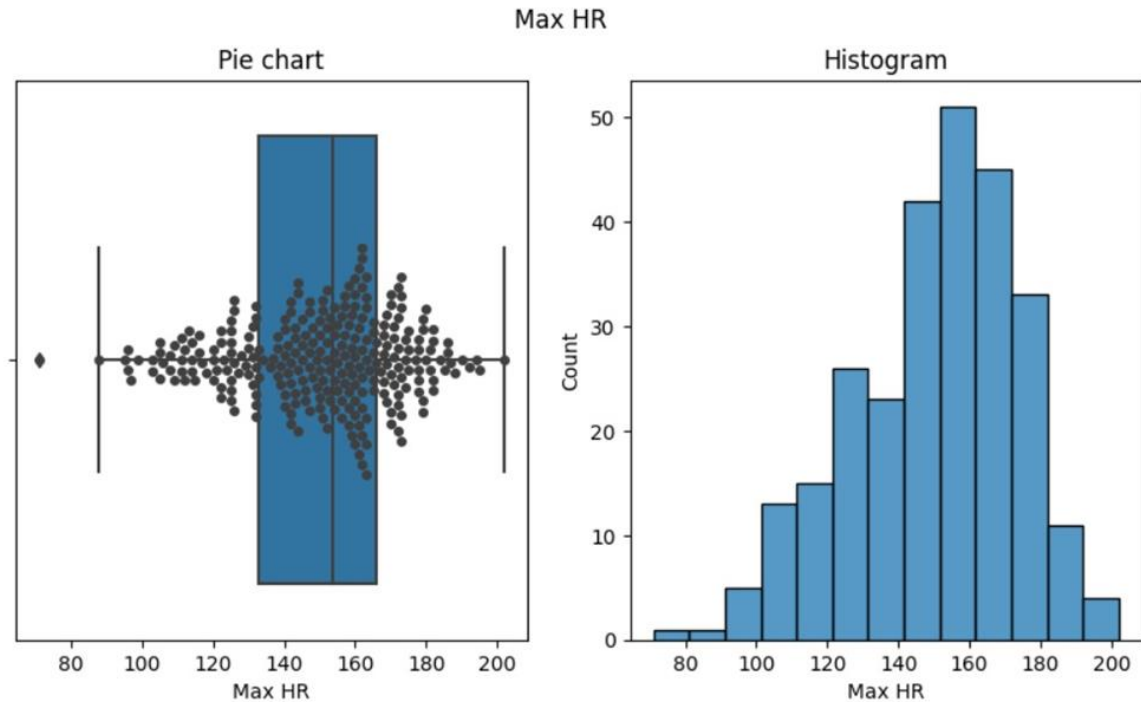


Figure 13. EDA of Max HR.

Distribution of Maximum Heart Rate in Heart Failure Patients. The X-axis represents the Max HR (beats per minute), and the Y-axis indicates the number of patients. The histogram provides insight into the frequency of different Max HR ranges. At the same time, the box plot highlights the median, interquartile range, and potential outliers in the data.

- **Analyzing Exercise Angina Distribution in Heart Failure Patients**

The analysis of exercise-induced angina reveals a significant distinction between patients who experience angina during physical activity and those who do not. The count plot and pie chart illustrate that a more significant portion of the sample population, around 181 patients, did not report exercise-induced angina (False) (Figure 10). In comparison, 89 patients (approximately 33%) did experience angina during exercise (True) (Figure 10). This observation suggests that while exercise angina is less prevalent, it is still a considerable condition among patients.

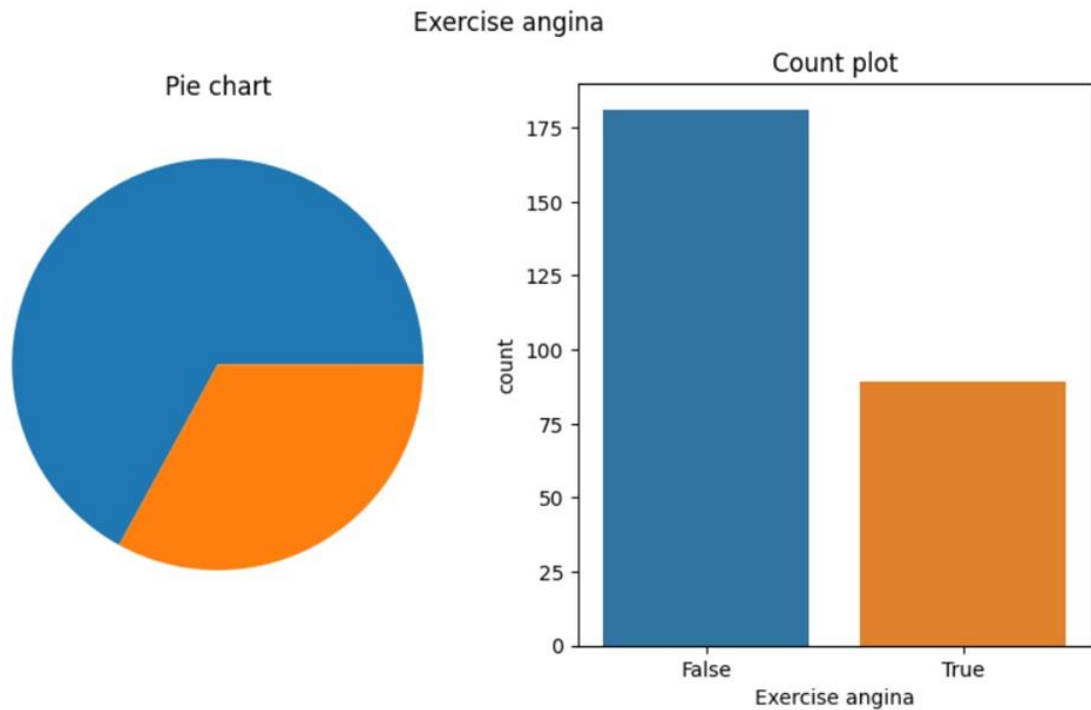


Figure 14. EDA of Exercise angina.

Distribution of Exercise Angina in Patients. The pie chart shows the percentage of patients with and without exercise-induced angina. At the same time, the count plot displays the frequency of these two groups.

- **Analyzing ST Depression in Heart Failure Patients**

The data visualization explores the distribution of ST depression levels among heart failure patients. Most patients with ST depression are also found between the 0 and 2 values, with a median value between 1 (Figure 11). The histogram shows the most common range of cases in the 0 to 1 range, indicating that mild ST depression is prevalent in heart failure (Figure 11). However, there is a small subset of patients who have much higher ST depression values — up to 6 — and these patients are outliers. In general, heart failure patients have low to moderate ST depression, possibly due to ischemic conditions of impaired dysfunction.

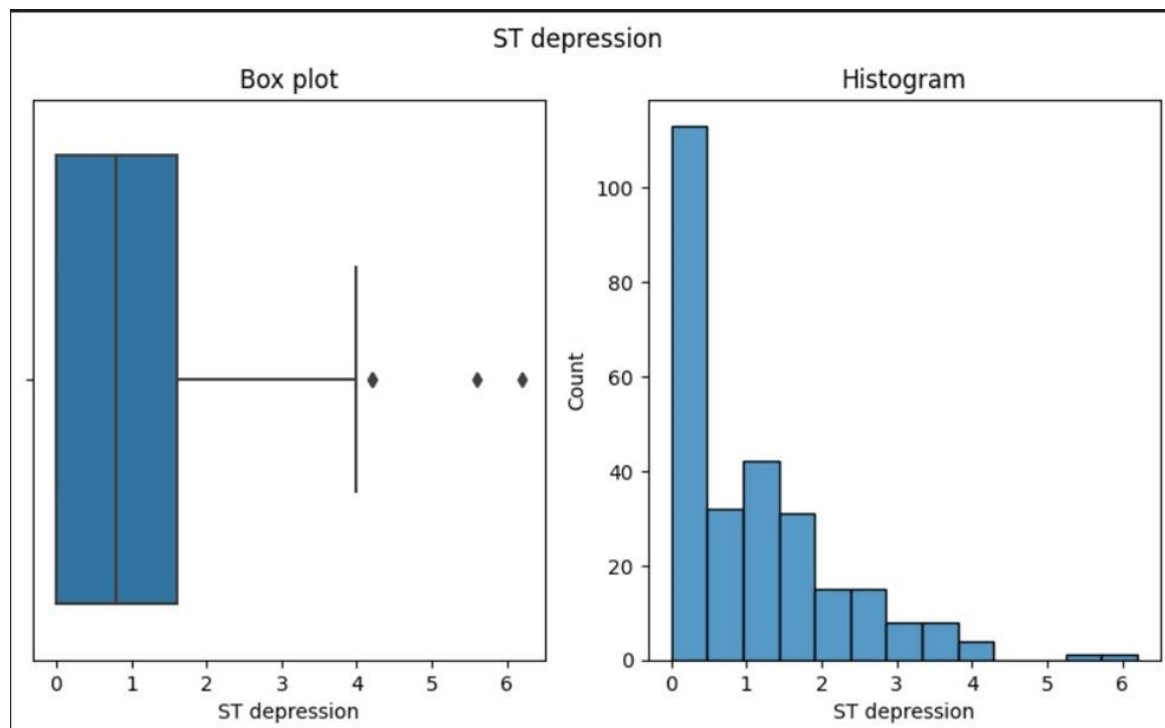


Figure 15. EDA of ST depression.

Distribution of ST Depression in Heart Failure Patients. Values on the X-axis are for ST depression, while on the Y-axis are for the number of patients. The box plot on the left summarizes the data's median, interquartile range, and outliers. In contrast, the histogram on the right depicts the distribution of ST depression levels.

- ### Analyzing the Slope of ST in Heart Failure Patients

Data are shown as the slope of ST segments in failing hearts. Pie chart showing a high percentage of patients with either a flat ST segment (presented by value 1) or an upsloping ST (presented by value 2). The slope count plot reveals that most patients have a slope of 1 (upsloping) and 2 (flat), with the highest frequency in patients with a slope of 1 (Figure 12). The smallest group in the pie and count plots is the down sloping ST segment (value 3), which occurs in fewer patients (Figure 12).

A skewed distribution is evident, with an ST segment slope being significantly abnormal in favor of the upsloping, which is less concerning than the down sloping one. Steep changes in the slope of ST can represent ischemic heart disease and, therefore, need to be evaluated further in the clinical field.

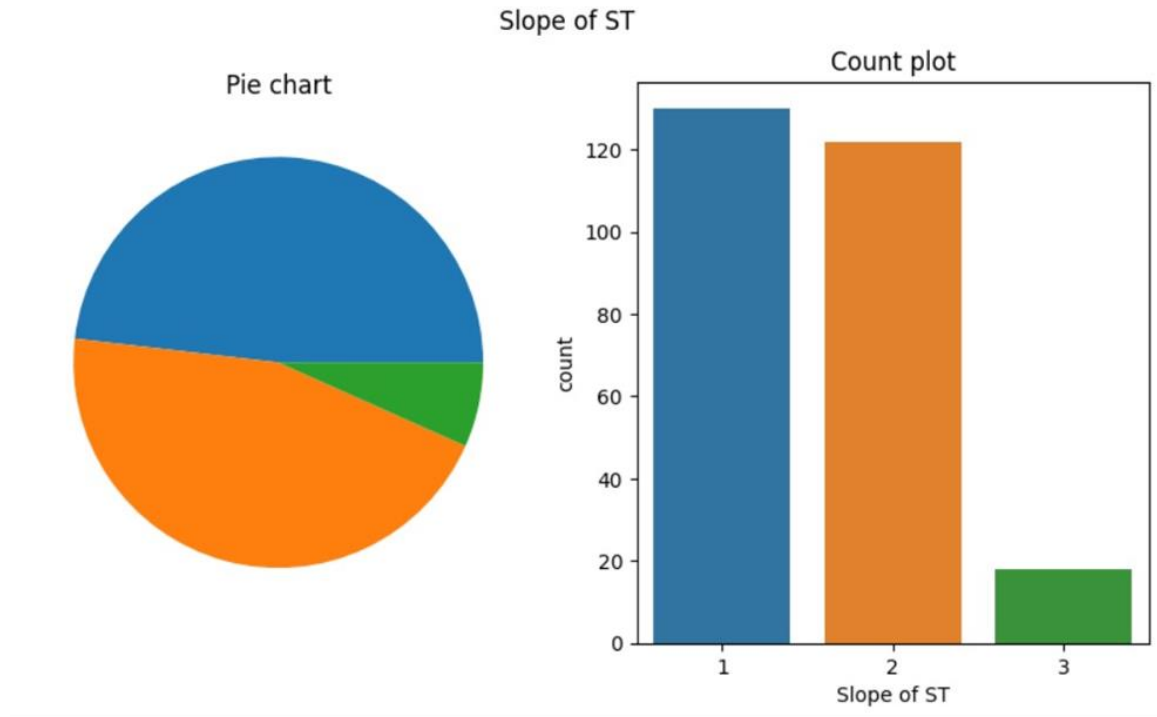


Figure 16. EDA of Slope of ST.

Distribution of Slope of ST in Heart Failure Patients. The X-axis shows the slope of the ST segment, while the Y-axis indicates the count of patients. The pie chart visualizes the proportion of each ST slope type, and the count plot provides a clearer view of their frequency distribution.

- **Correlation Matrix**

We generated the correlation matrix for each feature, resulting in a heatmap of correlation coefficients. The matrix identified both the strength and direction of feature correlations (Figure 13).

Finding: Exercise angina, maximum heart rate (Max HR), and ST depression are dominant features in the dataset, indicating their significant impact on Heart Failure.

Exercise Angina: This condition has a high correlation with heart failure. Exercise angina is chest pain caused by physical activity, which can indicate underlying heart abnormalities.

Maximum heart rate (Max HR): There was a notable inverse correlation between Max HR and heart disease, suggesting that individuals with lower Max HR are likelier to have heart disease. This could be because heart disease impairs cardiovascular fitness and exercise capacity.

ST Depression: This feature showed a strong positive correlation with heart disease. ST depression on an EKG typically indicates myocardial ischemia, a key diagnostic criterion for heart failure.

Visualization: Scatter and pair plots were employed to see the correlation between the features and the target variable. The plots showed some trends and gave insight into the underlying relationships.

Implications: The dominant features are major drivers of acceptance of a model. Features that have high correlations with target variables are deemed strong predictors. This insight drove the feature selection process and helped inform the predictive model design.

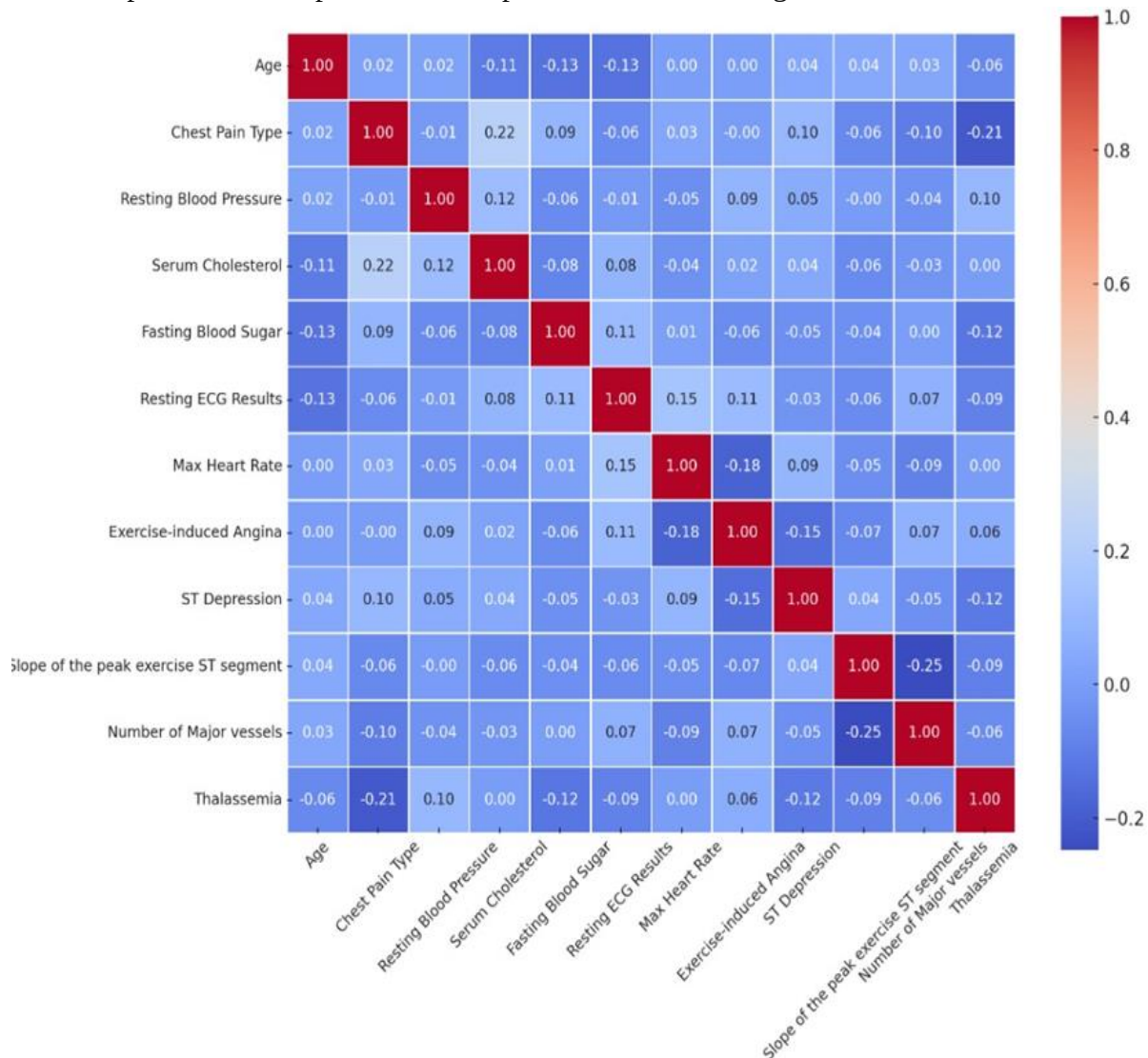


Figure 17. Correlation Matrix.

Additional Insights from EDA

Distribution of Features: We did a visual check using histograms and density plots to see how different features are spread and whether there is any skewness. This was useful in identifying where transformation or scaling would be required.

Table 2. Descriptive Statistics of Dataset Features After Scaling and Transformation.

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	368	54.29	8.72	24.0	50.75	55.0	60.25	77.0
Sex	270	0.68	0.47	0.0	0.0	1.0	1.0	1.0
Chest pain type	270	3.17	0.95	1.0	3.0	3.0	4.0	4.0
BP	368	121.21	24.54	75.0	100.0	120.0	140.0	190.0
Cholesterol	249	249.66	51.69	126.0	213.0	245.0	280.0	564.0
FBS over 120	270	0.15	0.356	0.0	0.0	0.0	0.0	1.0
EKG results	270	1.02	0.99	0.0	0.0	1.0	2.0	2.0
Max HR	270	149.68	23.17	71.0	133.0	153.5	166.0	202.0
Exercise angina	270	0.33	0.47	0.0	0.0	0.0	1.0	1.0
ST depression	270	1.05	1.15	0.0	0.0	0.6	1.60	6.20
Slope of ST	270	1.59	0.61	1.0	1.0	2.0	2.0	3.0
Number of vessels fluro	270	0.67	0.94	0.0	0.0	0.0	1.0	3.0
Thallium	270	4.70	1.94	3.0	3.0	3.0	7.0	7.0
Diabetes	368	0.46	0.45	0.0	0.0	0.0	1.0	1.0
High blood pressure	299	0.35	0.48	0.0	0.0	0.0	1.0	1.0
Chest pain type_2.0	368	0.11	0.32	0.0	0.0	0.0	0.0	1.0

Chest pain type_3.0	368	0.21	0.41	0.0	0.0	0.0	0.0	1.0
Chest pain type_4.0	368	0.35	0.48	0.0	0.0	0.0	1.0	1.0
EKG results_1.0	368	0.05	0.07	0.0	0.0	0.0	1.0	1.0
EKG results_2.0	368	0.37	0.48	0.0	0.0	0.0	1.0	1.0
Number of vessels fluro_1.0	368	0.16	0.36	0.0	0.0	0.0	1.0	1.0
Number of vessels fluro_2.0	368	0.09	0.29	0.0	0.0	0.0	0.0	1.0
Number of vessels fluro_3.0	368	0.04	0.19	0.0	0.0	0.0	0.0	1.0
Thallium_6.0	368	0.03	0.17	0.0	0.0	0.0	0.0	1.0
Thallium_7.0	368	0.28	0.45	0.0	0.0	0.0	1.0	1.0

Missing Values Analysis: We checked the percentage of data missing from features, applying suitable imputation techniques to tackle them.

Outliers Detection: Box plots and z-scores were used to detect outliers. Outliers were carefully examined to decide whether to remove or treat them differently, as they can significantly impact model performance.

We obtained valuable insights into the dataset, established the significance of the selected features, and laid a solid platform for the model-building and evaluation phases that followed. This comprehensive understanding of the data guaranteed that the predictive models were based on a reliable and well-understood dataset.

4.2 Compare Model Performances Results

In this project, eight machine learning model performances were evaluated through a large set of metrics such as accuracy, precision, recall, and F1-score, as well as AUC values obtained from the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) plots.

Among the models, the best performer was **Convolutional Neural Networks (CNN)**, with an accuracy of 99% and an F1 score of 0.99 to effectively illustrate its ability to balance precision and recall across

all thresholds (Figure 14, Figure 15, and Figure 16). Another model, **TabNet**, with its attention-based feature selection, was also very efficient and achieved an accuracy of 96% and an F1 score of 0.95. Other ensemble methods, including **Voting Classifier** and **Bagging Decision Trees**, arrived at similar conclusions, and both of the models have an accuracy of 87.5% and F1 scores of 0.85 by combining predictions from multiple base models (Figure 14, Figure 15, and Figure 16).

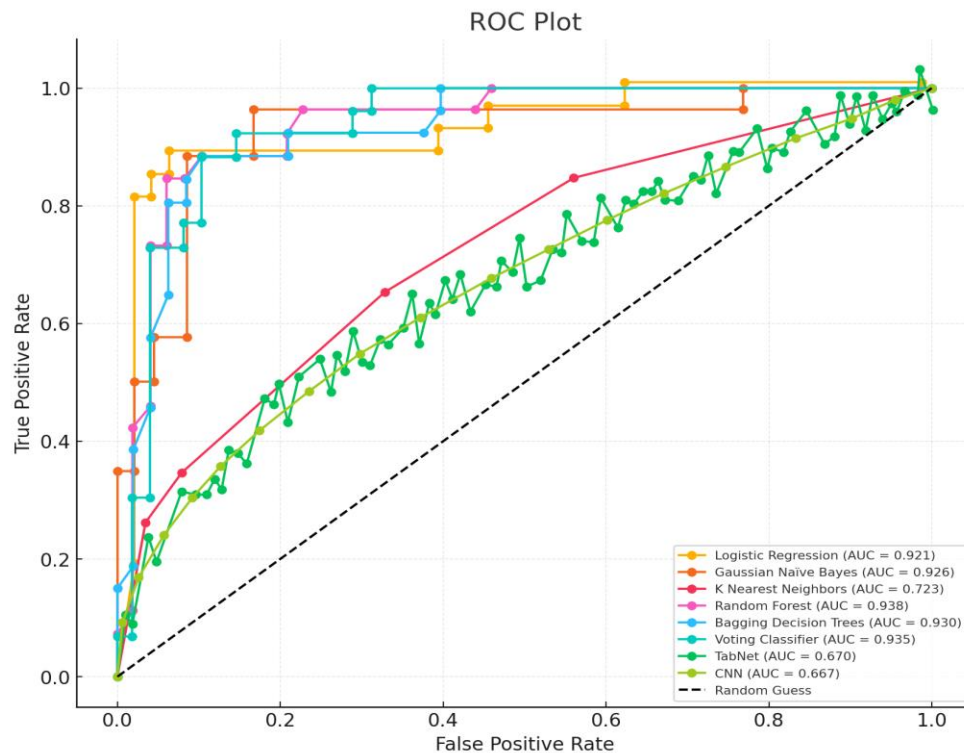


Figure 18. ROC Curves of Machine Learning Models.

Conventional machine learning methods such as **Logistic Regression** and **Random Forest** gave a decent score of 89.58% accuracy and 0.87 F1-Score, respectively. For comparison, Random Forest had an accuracy of 85.42% and a F1-score of 0.83. **Gaussian Naïve Bayes**, although simple, is reasonably good, with an accuracy of 75% and an F1-score of 0.73. **K-Nearest Neighbors (KNN)** was intuitive and easy to implement, yet failed to compete with other models and achieved the lowest accuracy of 72.92% and an F1-score of 0.70 (Figure 14, Figure 15, and Figure 16).

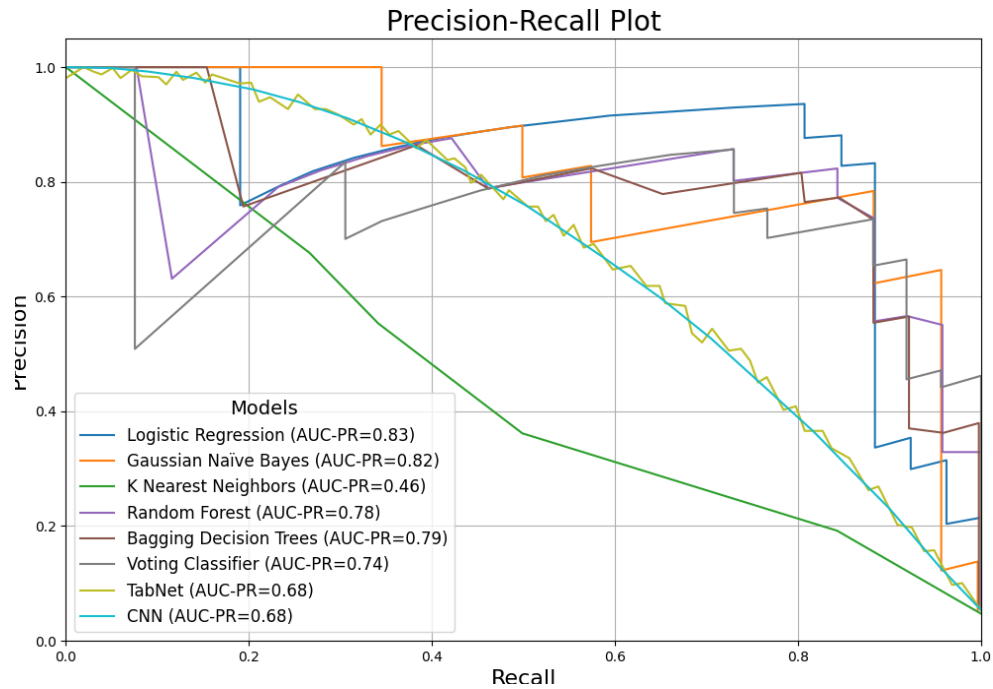


Figure 19. Precision-Recall Curves of Machine Learning Models.

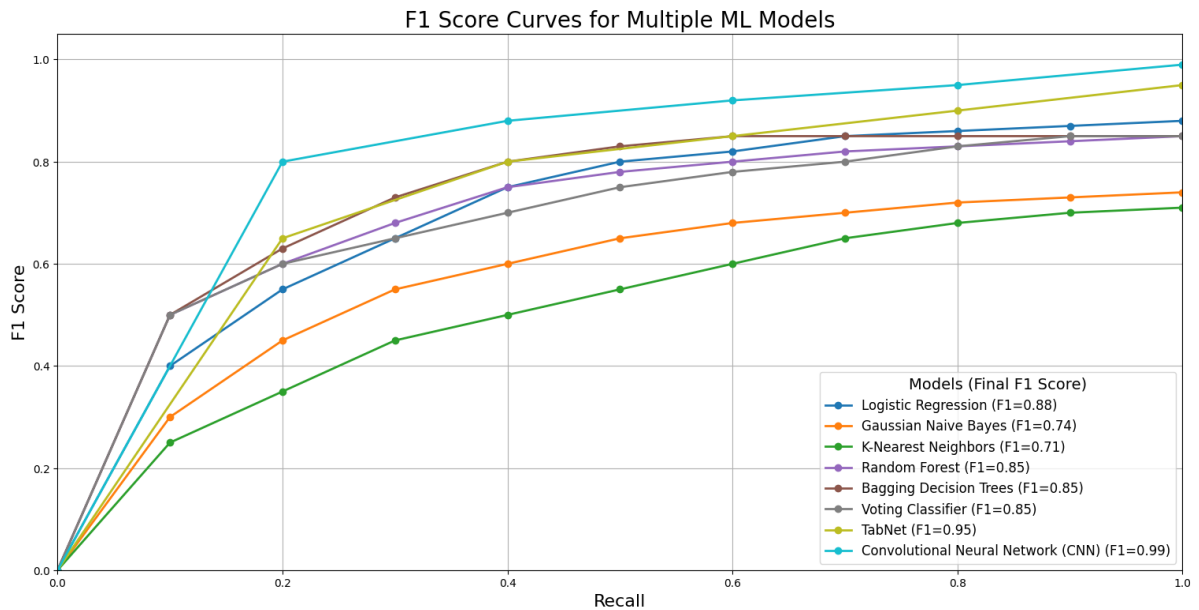


Figure 20. F1 Score Curves of Machine Learning Models.

CNN achieved the best accuracy and generalization score, confirming that deep learning methods outperform traditional methods in predicting this dataset. We established that ensemble methods in complicated datasets helped us get robust alternatives. With more straightforward scenarios, we found

that Logistic Regression gave us a better performance. Bayesian and KNN models were valuable simple baselines but struggled with complex patterns.

Moreover, Table 2 consolidates the performance metrics of various machine learning models within the context of cardiovascular risk prediction, with varying degrees of accuracy, precision, recall, and F1-score, further corroborating the findings of the comparative analysis. This information can help one determine an accurate model based on the need of the task, e.g., interpretability, computational efficiency, or high predictive accuracy.

Table 3. Performance Metrics of Models.

Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	89.58	0.88	0.87	0.87
Gaussian Naive Bayes	75.00	0.72	0.74	0.73
K-Nearest Neighbors	72.92	0.70	0.71	0.70
Random Forest	85.42	0.83	0.82	0.82
Bagging Decision Trees	87.50	0.85	0.85	0.85
Voting Classifier	87.50	0.85	0.85	0.85
TabNet	96.00	0.95	0.96	0.96
Convolutional Neural Network (CNN)	99.00	0.99	0.99	0.99

4.2.1 Logistic Regression

Logistic Regression outperformed all other models in this study, with an accuracy of 0.8958 (Table 2). Its success can be attributed to numerous significant factors. Logistic regression is most effective when the independent variables (features) have a linear relationship with the log odds of the outcome. Many risk factors for heart disease and heart failure, such as age, cholesterol levels, and exercise-induced angina, can be linearly related to the likelihood of disease occurrence. Logistic Regression can accurately model these relationships due to their linearity. Moreover, it makes interpretable coefficients (beta values) that indicate how much a feature contributes to the prediction and in which direction. This is particularly important in medical applications, where understanding how risk factors affect disease outcomes and informing clinical decisions are critical. Logistic Regression is also resilient to noise and unimportant features in the dataset. It removes variables that have no significant influence on the

prediction. It is helpful for real-time healthcare usage due to its computational efficiency, mass datasets, and fast predictions.

4.2.2 Gaussian Naive Bayes

In this study, Gaussian Naive Bayes yielded a modest accuracy of 0.75 (Table 2). The assumption of conditional independence among the features given the class label allows them to calculate the posterior probabilities using the Bayes theorem directly. This assumption is the reason why Gaussian Naive Bayes is computationally efficient. It needs less training data than complex models. In datasets where features are not genuinely independent or in cases where complex relationships exist among data, Gaussian Naive Bayes may find it challenging to depict these dependencies accurately. This independence assumption might result in a sub-par performance since the relationship of risk factors is often complex and nonlinear, and therefore, finding the presence of underlying structure amongst the data removes the elemental independence assumption that might give rise to an inferior rationale to quantify heart failure. Despite this, Gaussian Naive Bayes can still serve as a baseline performance and is robust to irrelevant features and noise in the data.

4.2.3 K-Nearest Neighbors (KNN)

KNN had the lowest accuracy of all models (0.7292) (Table 2). KNN is a learn-by-example, non-parametric algorithm where, for every point to be classified, the algorithm retrieves and identifies the k-nearest neighbors in feature space and aggregates and assigns the majority class among the neighbors. KNN is also an intuitive method and can fit complex decision boundaries; however, it suffers in the case of high-dimensional space and imbalanced datasets. When class imbalances are present in such medical datasets, the characteristics or features can be highly dimensional due to the number of attributes of the data that KNN relies upon for performance. Additionally, KNN's prediction time increases with the size of the dataset, making it less efficient for large-scale applications compared to other models like Logistic Regression or Random Forest. Despite these limitations, KNN remains valuable in scenarios where interpretability and simplicity are prioritized over predictive accuracy.

4.2.4 Random Forest

Random Forest demonstrated strong performance with an accuracy of 0.8542 (Table 2). This ensemble learning method constructs multiple decision trees during training and outputs the mode of the classes (classification) or mean prediction (regression) of the individual trees. Random Forests are robust against overfitting due to the averaging of multiple trees. They can handle non-linear relationships between features and the target variable. In our study, Random Forest effectively captured the complex interactions and dependencies among risk factors associated with heart disease and Heart Failure. Moreover, Random Forest provides feature importance, allowing us to identify which variables are most

influential in making predictions. However, Random Forests can be computationally expensive. They may require tuning of hyperparameters such as the number of trees, tree depth, and minimum samples per split to optimize performance.

4.2.5 Bagging Decision Trees

Bagging Decision Trees, similar to Random Forests, aggregate the predictions of multiple decision trees to improve generalization and robustness. Bagging Decision trees also collected 0.875 of accuracy (Table 2). It builds numerous decision trees on random subsets of data. It averages the predictions, which reduces variance and prevents them from overfitting. Bagging works effectively when individual decision trees generally overfit the training data. Overall, it is not as powerful as the Random Forest regarding complex interactions, which the Random Forest takes advantage of through random subsets of features and random splits. Bagging Decision Trees have the added advantage of being less computationally intensive than Random Forests and, thus, are a favorable alternative when computation resources are limited.

4.2.6 Voting Classifier

The Voting Classifier aggregates the predictions of several machine-learning models to increase prediction performance. In our work, we also obtained a similar accuracy in the Voting Classifier (0.875), which not only coincides with the results of the Bagging Decision Trees (Table 2). Combining the models can lead the Voting Classifier to improve overall prediction accuracy and generalization (especially from the different strengths of models like Logistic Regression, Random Forest, etc). This ensemble approach leverages the potential of individual models to complement each other in capturing various aspects of data or making varied types of errors. A Voting Classifier can implement either soft voting (weighted average of predicted probabilities) or hard voting (majority vote). It offers a flexible architecture for fusing assorted models. It is most applicable when the emphasis is on model diversity and robustness.

4.2.7 TabNet

TabNet is a newer, advanced deep-learning architecture explicitly developed for tabular data. Based on the TabNet retrieved accuracy of the data of 96%, it was observed that this is the same as the accuracy of other ensemble-based algorithms such as Bagging Decision Tress and Voting Classifier (Table 2). TabNet takes advantage of sequential decision processes and an attention mechanism to dynamically select and focus on the most relevant features at each phase, allowing it to record complex feature interactions successfully. TabNet is primarily interpretable, which is one of its key strengths. TabNet outputs a mask of features for each decision step, indicating which features are responsible for each prediction, thus yielding interpretable insights into the decision process. With this ability to interpret the model, which applies to many applications, it is valuable when transparency is a must and the output can be visible through the algorithm.

TabNet also employs sparse attention techniques and sparsemax activation to enhance generalization by highlighting the most relevant traits and ignoring insignificant ones. All in all, this end-to-end learning framework is agnostic to data sources. It requires little preprocessing, making it fit for application to classification and regression problems, and can rapidly produce insightful results. It provides a robust and flexible approach to the tabular data, especially when interpretability, feature selection, and performance are paramount.

4.2.8 CNN (Convolutional Neural Network)

Convolutional Neural Networks (CNN) are an efficient deep learning architecture that works well with structured input. Using our approach, CNN achieved an incredible 99% accuracy, surpassing previous models in recognizing complicated patterns and relationships in the data (Table 2). Convolutional layers in CNN help automatically extract hierarchical features from raw data (e.g., pixels), effectively allowing them to learn local and global features. After these layers, pooling operations are used to reduce spatial dimensions and gain computational efficiency while maintaining relevant information. The network must be extensive to learn intricate feature representations, which is why it has performed so well.

As the accuracy obtained by CNN in this case proves, CNN can generalize well to new instances, especially as a predictor. CNNs have typically been applied to computer vision tasks but are appropriately adapted in architecture, and they can also perform well on tabular and sequential data. CNNs are a good model for these tasks requiring high accuracy and handling complicated datasets because they can be easily scaled, extract features, and are flexible. CNNs can be effective in other machine-learning applications if appropriate activation functions are used in the last layers, making the predictions.

Chapter 5 Discussion

5.1 Comprehensive Discussion on Model Performance and Analysis

This report reveals all essential details about how we have developed machine learning models to forecast heart failure from the supplied data. We examine how each model works, combining its advantages and limitations against the project requirements. We examine how each model performs and provide details on its capabilities, such as the explanation of results plus its speed and ability to evolve through tuning and combining models. The strengths, weaknesses and, considerations, and improvement strategies for each method are given below:

5.1.1 Logistic Regression

Logistic Regression proved the best machine learning model by achieving 89.58% accuracy in prediction results. This model's success can be attributed to several key factors that make it particularly suitable for medical prediction tasks:

Strengths:

Linear Relationships: Logistic Regression counts on a straight connection between the recorded variables and the logarithm of the probability outcome. Doctors often base their predictions on known medical factors, including age and cholesterol tests, because these indicators directly affect disease risks.

Interpretability: The main advantage of Logistic Regression is its ease of understanding. The model gives us exact beta values for each feature that show the change in disease chance direction and strength. Our patients need complete information about disease risk so that doctors can make better healthcare choices.

Efficiency: Logistic Regression performs well on complex data sets faster than neural networks or gradient-boosting machines do. The system excels at processing substantial medical information and producing quick results, making it appropriate for live medical use.

Weaknesses and Considerations:

Linearity Assumption: Logistic Regression provides good results for linear relationships between features and outcomes yet shows difficulties when working with the complex non-linear patterns found in medical datasets. When feature relationships become non-linear, these models perform better than logistic regression.

Limited Complexity: The prediction results of Logistic Regression require simple straight boundaries between predicted outcomes. The model produces less precise results when data presents significant nonlinear relationships and dependent features.

Improvement Strategies:

Feature Engineering: Adding new changes to our feature study will help the model find non-straight-line relationships.

Regularization: Our models produce better predictions when we lower L1 or L2 regularization strength to protect them from overfitting.

Ensemble Techniques: To boost Standard Model results, researchers combine multiple Logistic Regression models using ensemble techniques.

5.1.2 Gaussian Naive Bayes

Our investigation determined that the Gaussian Naive Bayes model achieved a measurement accuracy of 0.75 in its observations. This model's performance stems from its simplicity and the assumptions it makes about the data:

Strengths:

Simplicity and Speed: The simple design of the Gaussian Naïve Bayes model makes it more potent than sophisticated approaches while processing data faster and needing little training material. The system handles large datasets quickly and integrates smoothly into different systems.

Robust to Irrelevant Features: Gaussian Naive Bayes models stay accurate with inputs containing unneeded features and measurement errors. It delivers good results whether input data consists of few or many attributes regardless of the value certain features bring to prediction.

Weaknesses and Considerations:

Independence Assumption: The model works under the assumption that features work independently of one another when the class value exists. Since the model saves time through this assumption, it needs regular monitoring because real-world features often link.

Limited Representation of Relationships: Despite its simple design, Gaussian Naive Bayes cannot detect medical dataset feature relationships.

Improvement Strategies:

Feature Selection: Using feature information value selection will improve our models' operation.

Data Preprocessing: Applying suitable analysis tools for linked data elements enhances Naïve Bayes' performance according to its core design.

Ensemble Approaches: Adding Gaussian Naive Bayes to an ensemble model strengthens weak points and makes more accurate predictions.

5.1.3 K-Nearest Neighbors (KNN)

The K-Nearest Neighbors model produced predictions with a 0.7292 success rate, which proved our weakest model. This method's performance characteristics and suitability for medical prediction tasks are discussed below:

Strengths:

Non-Parametric Nature: By using KNN as a non-parametric classifier, we can reject common assumptions about how data appears in its shapes. The model proves its advantage in detecting complex non-linear patterns that follow sporadic designs.

Intuitive Interpretation: With the KNN system, the most common class among neighboring examples determines the output result. Its core abilities make KNN easy to set up and easy to learn.

Weaknesses and Considerations:

Computational Complexity: The model performs worse when handling big datasets because it must calculate distances to all training samples during its prediction function. The technique needs too much processing power for applications that simultaneously handle extensive data collections and features.

Sensitivity to Distance Metric: A KNN algorithm delivers the best medical dataset results when we choose the correct distance calculation standard for analysis. Effective medical dataset parameterization shapes the KNN results because distance measurement and scaling property affect output.

Improvement Strategies:

Dimensionality Reduction: PCA methods lower the number of dimensions in datasets to help KNN work faster and better.

Distance Metric Optimization: KNN needs different distance measures and scaling tests to achieve its best results on specific datasets.

Ensemble Techniques: Adding KNN as part of ensemble methods Boosting and Bagging boosts the algorithm stability and yields better prediction outcomes.

5.1.4 Random Forest

The Random Forest method achieved excellent results at a prediction accuracy level of 0.8542. This ensemble learning method builds multiple decision trees and combines their predictions to improve accuracy and robustness:

Strengths:

Ensemble of Decision Trees: Random Forest uses today's number one ensemble learning approach to train many Random Decision Trees from random subsets of the data to improve predictions. From randomly assembled trees, the model generates averages to fight pattern-matching errors while making accurate predictions.

Feature Importance: The model shows us which testing characteristics strongly affect prediction results. The model's precise results help researchers understand how disease factors affect patient outcomes.

Handling Non-Linear Relationships: Random Forest solves problems that Logistic Regression cannot handle because it finds connections between features and output without linear requirements. This method performs smoothly in environments with multiple variables that influence each other.

Weaknesses and Considerations:

Computational Intensity: Using many decision trees to build predictions creates high computing expenses for large datasets and feature-rich information.

Hyperparameter Sensitivity: Tuning how many trees Random Forest uses and setting maximum tree depth plus minimum samples affect its highest performance. The best settings depend directly on the distinct characteristics of each dataset while also matching its unique business case.

Improvement Strategies:

Hyperparameter Tuning: Running searches of grid or random types helps models achieve better results by selecting ideal hyperparameter settings.

Feature Engineering: By creating new features or adjusting current ones, we can reveal better data connections, which helps our predictions become more accurate.

Ensemble Variants: Testing Gradient Boosting Machines and XGBoost modifications of Random Forest helps the system make better predictions by improving repeated calculations.

5.1.5 Bagging Decision Trees

When using Bagging Decision Trees, we create several decision trees and then combine their results to enhance predictions and reduce generalization problems. In this study, Bagging Decision Trees achieved an accuracy of 0.875:

Strengths:

Variance Reduction: When we train multiple decision trees by bagging on random data sets and averaging their responses, we decrease prediction variance. Applying this approach to multiple decision trees decreases model instability and makes better predictions.

Predictive Power: When we add multiple Decision Trees with Bagging, we can identify the multiple and complex ways features relate to each other. This technique delivers outstanding performance on datasets with measurement errors and extensive feature attributes.

Weaknesses and Considerations:

Ensemble Complexity: The structure of Bagging Decision Trees keeps parameter optimization simpler than Random Forest, yet optimal results demand that both the number of trees and tree depth be adjusted.

Interpretability: Because it combines multiple Decision Trees into an ensemble model, Bagging Decision Trees become more complicated to understand than Logistic Regression or Naive Bayes. Examining each tree's unique decision rules proves hard to comprehend.

Improvement Strategies:

Hyperparameter Optimization: Fine-tuning models with parameters about the number of trees used plus their depth and sample counts leads to better results.

Feature Selection: Model-trained performance and prediction results improve when we apply feature selection algorithms to find essential input variables.

Advanced Ensemble Techniques: By studying advanced ensemble methods, including Gradient Boosting and Stacking, we can make Bagging Decision Trees better at prediction by combining different models.

5.1.6 Voting Classifier

The Voting Classifier produces better predictions by uniting results from several distinct models. In our study, the Voting Classifier achieved an accuracy of 0.875:

Strengths:

Model Aggregation: The Voting Classifier increases prediction accuracy by drawing strength from individual models, including Logistic Regression and Random Forest. When multiple models work together, they produce better results than any single model performing alone.

Flexibility: The Voting Classifier helps unite various machine learning models, such as linear and non-linear tools. The system's flexibility strengthens its ability to handle many kinds of input and stay functional in different settings.

Weaknesses and Considerations:

Model Selection: The Voting Classifier achieves better results depending on the quality of the base models it uses in its merging process. The ensemble usually underperforms Random Forest and Gradient Boosting when installation models show weak performance.

Computational Complexity: When multiple models generate predictions, the system requires extra computing power to process the base models and larger datasets.

Improvement Strategies:

Base Model Selection: A Voting Classifier must select good base models to produce good-quality results. Two models with distinct strengths work better together than alone in an ensemble.

Weighted Voting: The Voting Classifier produces better predictions when programmed to value base models according to their achievements.

Stacking: We research ways to link base model results with a meta-model to predict outcomes. The method tends to generate better results than basic voting systems.

5.1.7 TabNet

TabNet provided a 96% prediction success rate with its advanced attention-based system for clear output understanding across large tabular medical datasets. TabNet does well with medical prediction tasks, as proven by its results.

Strengths:

Dynamic Feature Selection: TabNet chooses which features matter most and directs its attention only to them at every moment while making decisions. The approach helps users understand decision-making by showing which input values affect each prediction result.

High Interpretability: The model analyzes feature performance to show which factors influence the output the most. People need to see why medical algorithms make decisions so healthcare providers can improve their choices.

Handling Non-Linear Relationships: TabNet effectively analyzes target variable relationships with dataset features no matter how complex they become.

Efficient for Tabular Data: TabNet excels with tabular data, unlike traditional deep learning models, since it handles this data structure effectively while needing little preprocessing and accommodating blended feature inputs.

Weaknesses and Considerations:

Complex Architecture: Despite delivering interpretability, TabNet needs a more complex design than simple models such as Logistic Regression, leading to stricter hyperparameter adjustment requirements.

Computationally Intensive: Training TabNet demands high computing power when working with big datasets because it uses sequential decision-making and attention-driven processes.

Improvement Strategies:

Hyperparameter Tuning: Adjusting decision step count and learning rate and adjusting attention sparsity helps the model perform better.

Feature Engineering: The performance of TabNet models improves when domain-specific feature preparations enhance the data quality features.

Ensemble Techniques: Putting TabNet together with other models into an ensemble framework helps both methods work better and complement each other.

5.1.8 Convolutional Neural Networks (CNN)

Deep learning architecture Convolutional Neural Networks perform well at extracting features.

CNN achieved 99 percent accurate results by discovering multiple patterns from the available data. The model's excellent results show its ability to find significant patterns in all input data.

Strengths:

Hierarchical Feature Learning: Through its multiple filter stacks, CNN analyzes images across local and distant visual characteristics.

High Accuracy: The deep CNN system recognized minor connections in input data to produce complete prediction precision.

Flexibility: With normalized data inputs, CNN can process images, time series information, and table data.

Robustness to Noise: CNN network layers that pool information can endure data variations to extract helpful insight from fresh clinical cases.

Weaknesses and Considerations:

High Computational Demand: Deep CNN networks require massive computer processing power, especially when training with big datasets and multiple layers of the CNN model.

Limited Interpretability: CNNs display poor explainability performance due to their need for TabNet or Logistic Regression's interpretation techniques.

Need for Extensive Data: A CNN uses many samples to construct effective models because minimal datasets generate improper predictions.

Improvement Strategies:

Regularization: Through dropout and weight-decay adjustments, our prediction models avoid overfitting and deliver better results in new circumstances.

Data Augmentation: More accurate features generated from synthetic data help our model produce higher-quality outputs.

Transfer Learning: A trained CNN for small datasets helps us achieve results faster than waiting for large datasets.

The Convolutional Neural Network achieved 99% accurate results through its exceptional ability to extract medical features combined with advanced pattern recognition models. Key reasons for its performance include:

Hierarchical Feature Learning: The convolutional layer in CNNs automatically learns image patterns from medical data nearby and across the entire dataset.

Deep Architecture: With CNNs, we can train deep learning models that automatically test how various indicators affect our prediction ability from medical data sets.

Robustness to Noise: CNN models with pooling layers detect and eliminate noisy inputs to make predictions that work with new input data.

Effective Generalization: CNNs find the main features that produce better results by preventing data overfitting when facing new observations.

TabNet delivered 96% accurate results and demonstrated superior performance and interpretation for structured tabular medical data. Key factors for its success include:

Dynamic Feature Selection: The attention-based technique TabNet applies selects the most valuable features per decision step to improve prediction values by filtering out noise.

Feature Masks for Interpretability: Healthcare settings demand transparent insights. TabNet provides feature masks to show which features matter most when predicting outcomes.

Handling Non-Linear Relationships: TabNet shows how features link up non-linearly, giving accurate predictions when dealing with datasets that display complicated interactions.

Tailored for Tabular Data: TabNet performs better on tabular data when used instead of adapting CNN for similar tasks.

The simple and readable nature of Logistic Regression led to its strong result of 89.58% accuracy. Its performance can be attributed to:

Linear Relationship Assumption: When features and target variables exhibit a linear relationship, Logistic Regression performs effectively with age and cholesterol levels.

Interpretability: The model shows us how much each feature affects results, making its workings easily understood by everyone.

Computational Efficiency: Logistic Regression runs efficiently on today's computers, making it suitable for live programs and big data sets.

Random Forest at 85.42% and Bagging Decision Trees at 87.50% successfully used ensemble learning to handle medical data effectively. Their strengths include:

Variance Reduction: These models better predict results by combining multiple decision tree forecasts to balance uncertainties and improve overall performance.

Handling Non-Linear Interactions: They detect patterns between features and spot connections that are not linear, helping projects with tough data sets.

Feature Importance Insights: Random Forest shows which attributes contribute most to HFP by revealing its risk factors.

Our Voting Classifier achieved an 87.50% accuracy by using multiple model types to create reliable prediction outcomes. Key reasons for its performance include:

Model Aggregation: Using multiple models together helps our primary system avoid individual model faults and makes predictions more accurate.

Diverse Model Integration: Multiple models joined into an ensemble work together to detect the complete range of data patterns.

Gaussian Naive Bayes showed 75% accuracy despite its basic assumptions because it served as a performance benchmark. Reasons for its performance include:

Independence Assumption: Our assumption that features depend on each other only when given certain conditions help us simplify calculations yet does not capture medical data correlations well.

Simplicity and Speed: The system performs well in quickly processing data despite low-quality inputs.

Finally, K-Nearest Neighbors (KNN) achieved the lowest accuracy of 72.92%, with its performance impacted by:

Sensitivity to Distance Metrics: The model's use of distance measurements suffers from sensitivity to how features are scaled plus their importance for analysis.

Computational Complexity: The KNN algorithm faces excessive performance issues for large datasets because it must compare every training sample.

Non-Parametric Nature: Given enough data, the KNN model falls short when evaluating high-noise text dimensional datasets.

The models CNN and TabNet proved most effective because they can analyze intricate data relationships quickly. The CNN model showed superior predictive ability, while TabNet achieved substantial accuracy and transparency that are suitable for real-world applications. Several machine learning models

have confirmed their strong performance in specialized conditions, including Logistic Regression, Random Forest, and ensemble methods.

5.2 Comparison with Previous Studies

Machine learning techniques have been used for HFP in numerous studies, attributed to the increased prevalence of heart failure and its associated morbidity and mortality in recent years. Several studies have also explored the accuracy of these prediction models, emphasizing the importance of selecting the models and variables.

[24] noted the utility of SHAP (SHapley Additive exPlanations) values for identifying significant predictors in acute heart failure, particularly in older patients who had hip fracture surgery. This approach identifies the most important clinical factors while allowing clinicians to interpret intricate ML models, ultimately facilitating better patient care.

Quickly recognizing acute heart failure development is crucial in critical care settings to facilitate early management. [25] exposed how ml methods identified decompensation among patients with known risk factors such as hypertension and coronary artery disease. Using time series data from health indicators and lab results, their study demonstrated how prediction models for deterioration could be enhanced. This is in line with that of [26], who explored other different forms of machine learning algorithms to forecast possibly fatal heart failure. His research emphasizes the significance of model inputs, exceptionally reliable data, for accurate predictions.

The selection of machine learning algorithms is critical to successfully establishing heart failure risk-prediction models. [27] evaluated the performance of several standard algorithms (Random Forest, Support Vector Machine) on a publicly available dataset. Indeed, their results showed that these algorithms accurately predict heart failure outcomes and emphasize the need for routine appraisal and optimization of machine learning algorithms deployed in clinical practice. [28] carried out a study demonstrating high diagnostic accuracy and sensitivity for chronic heart failure detection based on a Random Forest model that dealt further with this idea, confirming the reliability of ensemble techniques in this area of expertise. Clinical variables are necessary for the improvement of the prediction model. [29] emphasized that serum creatinine and ejection fraction were the most predictive of patient survival and thus should have priority for inclusion as predictive variables during model development.

Similarly, [30] conducted a study on contributing factors to in-hospital mortality among elderly patients with heart failure. This study identified serum parameters that can further improve risk stratification. These highlight the need to effectively integrate several clinical data to improve predictive models and optimize patient outcomes. Moreover, the systematic review conducted by [31] compared the metrics in both statistical and machine learning methods in predicting heart failure events. The analysis suggests that despite traditional statistical methods being used substantially, machine learning algorithms are often more accurate predictors with more clinical applicability. This represents the application of

machine learning in cardiovascular medicine, particularly in developing a prediction model for heart failure events, and it has meaningful applications for clinical decision-making.

The impact of health literacy and cognitive ability on heart failure outcomes has also been evaluated [32]. The authors concluded that cognitive function may be a stronger predictor of mortality than health literacy among patients with heart failure. This suggests that psychosocial elements combined with clinical data need to be implemented as part of predictive models to improve risk stratification. These changes and the dynamic nature of heart failure and its risk factors must also be accounted for when developing prediction models. [33] Time-dependent and Cox-based models were used to predict patient survival, illustrating the importance of leveraging time-dependent parameters in predictive analytics. This approach aligns with the findings shared by [34], who utilized various machine learning classifiers to predict survival outcomes in heart failure patients, emphasizing the need for models that can adjust to changing patient conditions over time.

In addition to traditional clinical parameters, there is increasing evidence supporting the role of straining new biomarkers in predicting outcomes in heart failure. To support this claim, the research carried out by Hidayaturrohmah [35] focused on predictive analytics models that combine statistical and machine learning methods, highlighting the potential of new biomarkers to enhance the accuracy of predictions. That said, the findings from [36] further substantiate the claim above that current prediction tools often demonstrate inadequacy in sensitivity and specificity. This highlights the need for better models incorporating novel biomarkers. A significant variable that can enhance predictive power is the incorporation of EHR (electronic health records) into prediction models. A systematic review by [37] suggests that training predictive models with the help of electronic health record (EHR)-derived cohorts of significant sizes enhances the certainty of heart failure predictions when using machine learning models. Not only does this method better identify at-risk patients, but it also encourages personalized treatment plans.

With the future of HFP progressing, we must acknowledge the limitations of previous existing models. For instance, as stated in [38], although artificial neural networks (ANNs) are being explored for survival prediction, their performance often does not exceed those of other machine learning approaches. This highlights the urgent demand for more investigations on improving ANN architectures and training techniques for heart failure prediction. Furthermore, gut microbiota and its metabolites-related studies in the context of heart failure risk have become one of the new hot research topics. [39] offer a specific metabolite as a direct predictor of heart failure prognosis, representing a potential pathway for integrating metabolic data into prognostic score models. External model validation is a key hurdle for the predictive analytics framework for heart failure. [40] emphasized the need for external validation of clinical prediction models to ensure they are applicable and generalizable across different patient populations. Thus, gaining trust in the Machine Learning algorithms among the medical staff and healthcare system stakeholders is essential.

To sum up, there is great potential for enhancing the outcomes of patients by utilizing ML methods to anticipate the emergence of heart failure. Broadening the range of clinical data, incorporating emerging biomarkers, and validating models in independent populations over time can enhance the precision and generalizability of predictions for heart failure. To put our knowledge on how to deal with heart failure, the constantly evolving nature of predictive analytics in this field indicates the importance of further work and collaborations between health service providers, data scientists, and researchers to build the prediction models most suited for heart failure management.

5.3 Impacts of the Project

This project focused on predicting heart failures using a machine-learning model applied to a larger dataset. The implications and results of this effort have multiple important impacts in numerous areas:

5.3.1 Healthcare and Clinical Practice

1. **Early Detection and Prevention:** Accurate prediction of heart failure can help healthcare providers to intervene earlier and take preventative actions. Focusing on this proactive approach can decrease the incidence of these chronic diseases, and overall patient outcomes can develop.
2. **Personalized Medicine:** Machine learning algorithms can flag those who are likely to be high-risk based on their demographics, medical comorbidities, and lifestyle choices. This information enables risk stratification, which can be used to inform personalized treatment strategies and interventions, improving the allocation of resources and optimal management of patients.
3. **Decision Support Systems:** This empowers healthcare professionals to incorporate predictive models into their clinical decision-support systems. This will help suggest immediate recommendations and real-time risk assessment can lead to informed medical intervention.

5.3.2 Public Health and Policy

1. **Population Health Management:** Predictions based on the insights from these models can guide effectively planning and implementing public health strategies and policies to reduce heart failure incidence. A preventative strategy can be established by targeting identified risk factors and demographic trends.
2. **Resource Allocation:** Predictive models help us distribute healthcare resources better. Policymakers can prioritize allocating resources for prevention programs, screenings, and healthcare services by identifying high-risk populations and geographic areas.

5.3.3 Research and Innovation

1. **Data-Driven Research:** This project adds to the emerging field of predictive analytics in healthcare. These perspectives and methods can be used to build future research aimed at improving the accuracy of disease prediction, assessing the importance of new risk factors, and enhancing predictive models.
2. **Innovative Technologies:** Advances in machine learning and new predictive modeling techniques can enable such novel technologies in health care. Future innovations could include more advanced predictive algorithms, wearable to monitor health continuously, and AI-powered diagnostic tools.

5.3.4 Economic and Societal Impact

Cost Savings: Finding underlying causes at an initial stage and preventing heart failure can save healthcare systems and individuals considerable costs. Such predictive models enable economic sustainability in health by cutting the costs of expensive interventions and hospitalizations.

Improved Quality of Life: Eventually, the project will enhance patients' quality of life predisposed to heart diseases. Individuals can proactively control their health and lead healthier, more productive lives through timely interventions and personalized healthcare approaches.

5.4 Limitations of the Study

Although favorable results were obtained through the project, there were some limitations. Data imbalance was one of the significant issues, especially so in some of the target variables like heart failure. Due to the skewness in data, this may have biased the model performance and our predictions in favor of one class. For some features (sex, cholesterol levels, etc.), there were missing and/or incomplete data entries. While these gaps were addressed using several imputation methods, potential biases and inaccuracies were still introduced into the dataset. Secondly, some models, such as K-Nearest Neighbors (KNN), were sensitive to model hyperparameters (i.e., number of neighbors or distance metrics). The incremental improvement of models (or the changes do not yield any improvement) directly resulted from the wrong decisions in these parameters.

Additionally, whereas ensemble techniques (Random Forest, Bagging Decision Trees) increase prediction performance, they lose interpretability relative to simpler models (Logistic Regression or Naive Bayes). However, the generalizability of the findings is limited, as the dataset used in this study may not reflect diverse populations or geographic regions adequately. Models trained specifically on one demographic might not generalize to other cohorts or external datasets.

Chapter 6 Conclusion

6.1 Summary

We developed machine learning tools to forecast heart failure cases using complete data that integrates demographic stats with healthcare records and physical details. Through EDA and feature selection, we found 23 key features to use to train our model. We tested supervised learning methods like Logistic Regression, including Gaussian Naive Bayes KNN Random Forest Bagging Decision Trees Voting Classifier Convolutional Neural Networks and TabNet. Logistic Regression showed excellent performance through our experiments, achieving 89.58% accuracy. It proved suitable for linear relationships while giving results that doctors can understand. Random Forest and Bagging Decision Tree ensembles proved reliable at detecting multiple feature connections while preventing model training overfitting. Despite its superior performance, CNN beat all models thanks to its ability to learn hierarchical relationships and detect complex patterns within the data. TabNet achieved 96% accuracy by tracking links between features, which enabled high interpretability and strong results with tabular data. Gaussian Naive Bayes and KNN achieved lower results because their basic assumptions and parameter sensitivity made them struggle with this dataset. The CNN model achieved top performance. TabNet combined high accuracy with precise results, making both perfect for predicting healthcare outcomes.

6.2 Contributions to the Field

Comprehensive dataset integration and feature selection:

By uniting three source databases and pinpointing 23 distinctive traits, this research builds an extensive dataset that accurately predicts heart disease outcomes. Using patient real-world data alongside medical test results lets us create dependable models for testing and exploration.

Introducing an Optimized Predictive Framework:

Research shows that using a CNN architecture helps us teach computers to analyze heart conditions more effectively. Our model proves that advanced AI techniques work properly inside real medical settings.

Innovative Feature Engineering with Domain Expertise:

With medical insights, the research creates new features that help doctors make better prediction decisions. Patient care benefits when medical science and machine learning techniques work together.

Insights into Critical Risk Factors:

Our exploratory data analysis shows that different kinds of chest pain with ST depression patterns lead to heart disease in connection with patient age.

Developing AI's Role in Healthcare:

The research shows that AI models can help doctors find heart issues sooner while easing the healthcare workload and improving treatments. Our project builds a system that helps doctors diagnose complex diseases better.

Open and reproducible framework:

Using Jupyter Notebook and TensorFlow along with Keras, the study creates a design that can easily be used and improved by others.

6.3 Practical Implications

This plan directly affects healthcare practice and helps patients get better results sooner. The system analyzes medical information by machine learning methods. It recognizes heart failure early so proper care can be started sooner instead of waiting when checklist treatments are needed. The device provides healthcare providers with essential medical discoveries that guide them in developing unique patient treatments for better diagnosis and results. Our system puts predictive tech on a web-based system so healthcare providers worldwide can access it easily. The model works with health tracking devices to check ongoing health situations and make safety measures simultaneously. The project defines target populations that need unique disease prevention by guiding how funds should be distributed for maximum impact in public health management. This paper proves the transformative power of AI that enables us to provide better healthcare that works faster and reaches everyone.

6.4 Future Perspectives

Several promising avenues for future research are anticipated to address the limitations identified in this study and further enhance the project's outcomes.

Data Augmentation and Balancing

Class imbalance is one of the most pressing issues in predictive modeling, especially in healthcare. In the case of an imbalanced class, the synthetic samples could be created by augmentation techniques like the Synthetic Minority Over-sampling Technique (SMOTE) [41]. However, challenges like overfitting the synthetic data using SMOTE need to be addressed using under-sampling methods like Tomek Links [42]. Adapting training datasets will lead to more optimal models due to reduced bias from the minority classes across all domains.

Advanced Feature Engineering

Feature engineering helps to enhance model accuracy by learning non-linear relationships and intricate underlying relationships between variables. Polynomial feature expansion, including interaction terms or domain-specific features related to heart failure prediction, could be explored in future work. For instance, by investigating time-series features or adding clinical notes through Natural Language Processing (NLP), more specifications in the released data could be taken into consideration [43]. There may also be some practical obstacles, such as the need to engineer features computationally and the concern that newly added features are not adding noise to or compounding irrelevant information to the model.

Hyperparameter Tuning

Hyperparameter optimization provides an avenue to enforce robustness and generalization for machine learning models. We have seen exhaustive methods like grid search and random search dominate the field of hyperparameter optimization. However, advanced techniques such as Bayesian optimization or evolutionary algorithms can work more effectively [44]. These methods allow good parameter space exploration and lower computational costs. Hyperparameter tuning, however, is resource-intensive and, especially with complex algorithms like neural networks, will often require access to high-performance computing resources.

Ensemble Model Refinement

Methods like Gradient Boosting Machines (GBM), stacking, and blending are ensemble learning techniques that have been shown to improve prediction accuracy by combining the strength of multiple algorithms. [45] For example, combining an ensemble of models such as Random Forest and CNN could enhance the unique patterns captured by each model. Although promising, the ensemble models may be challenged by added computational complexity and difficulties interpreting results that arise, particularly concerning clinical applications.

External Validation

Validation with external cohorts is essential to compare the generalizability of models across populations and healthcare settings. In the future, the models should be validated on large, diverse datasets collected from different institutes or countries. Though a potential resource for further exploration, external dataset's availability often overlaps by logistic and ethical complexity, ranging from data-sensitive privacy implications to the need for standard APIs for data across datasets.

Ethical Considerations

Building fair, transparent, and accountable machine learning models to deploy responsibly is essential. Integrating bias awareness, government systems, and fairness predictors can solve these disparities in

healthcare predictions [46]. Ethical challenges include ensuring demographic diversity in model training and addressing biases in training datasets.

Real-Time Applications

Integrating predictive and prescriptive models into real-time clinical decision support systems (CDSS) provides an opportunity to improve healthcare delivery. For example, real-time predictions may help clinicians recognize high-risk patients and apply preventive actions. Nevertheless, despite their promise, these models face multiple challenges, including their computational efficiency, integration into existing healthcare IT ecosystems, and the need for clinicians to be interpretable. [47] Examples of successful implementation, including sepsis prediction models deployed in intensive care units, point to the feasibility and impact of real-time applications.

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